EE2004: Digital Signal Processing

Tutorial 1

(January 29, 2026)

1. Determine whether the following signals are energy signals, power signals, or neither: (a) $(-0.5)^n u[n]$, (b) $x[n] = u[n]$, (c) $2\,e^{j3n}$, (d) $x[n] = 2^n\,u[n]$. Note that $u[n]$ is the unit step sequence, which is defined as 1 for $n \geq 0$ and 0 for $n < 0$.

2. Let $x[n] = \{-1,\ 0,\ \underset{\downarrow}{1},\ 2,\ 5\}$ (where the arrow indicates the sample corresponding to the index $n = 0$). Sketch and label carefully each of the following signals: (a) $x[4-n]$, (b) $x[n]\,u[2-n]$, (c) $x[n-1]\,\delta[n-3]$, (d) $x[n^2]$, (e) even part of $x[2n+1]$.

3. Let $x[n]$ be a signal such that $x[n] = 0$ for $n \notin \{-2,-1,0,1,2,3,4\}$. For each signal given below determine the values of n for which it is guaranteed to be zero. (a) $x[n+4]$, (b) $x[-n]$, and (c) $x[-n+2]$.

4. Let $x[n] \in \mathbb{R}$. Define $x_e[n] = \dfrac{x[n]+x[-n]}{2}$ and $x_o[n] = \dfrac{x[n]-x[-n]}{2}$. Define $E_x = \sum_n |x[n]|^2$. It is claimed that $E_x = E_{x_e} + E_{x_o}$. Prove or disprove the claim.

5. Sketch the odd and even components of the following signals: (a) $x[n] = 1$ for $0 \leq n < 6$, and (b) $x[n] = \{\ldots, 0, \underset{\downarrow}{0}, 1, 2, 1, 0, \ldots\}$.

6. Find the odd and even components of: (a) $u[n]$, (b) $n\,u[n]$, (c) $\sin\left(\frac{n\pi}{4}\right)$, (d) $\cos\left(\frac{n\pi}{4}\right)$, (e) $\sin\left(\frac{n\pi}{4}\right)\,u[n]$, (f) $\cos\left(\frac{n\pi}{4}\right)\,u[n]$, (g) $\cos\left(\frac{n\pi}{4}+\frac{\pi}{4}\right)$, (h) $u[n+5]-u[n-5]$

7. Let $x[n] \in \mathbb{C}$. The *conjugate even part* of $x[n]$ is defined as

$$x_{ce}[n] = \frac{1}{2}\left(x[n] + x^*[-n]\right)$$

$x_{ce}[n]$ is also called as the *conjugate symmetric part*. Similarly, the *conjugate odd part* of $x[n]$ is defined as

$$x_{co}[n] = \frac{1}{2}\left(x[n] - x^*[-n]\right)$$

$x_{co}[n]$ is also called as the *conjugate anti-symmetric part*. Note that whenever $x[n] \in \mathbb{R}$, $x_{ce}[n] = x_e[n]$ and $x_{co}[n] = x_o[n]$.

Find the conjugate symmetric and conjugate anti-symmetric part of $x[n] = e^{jn}$.

8. Let $x_{ce}[n]$ and $x_{co}[n]$ be the conjugate-even and conjugate-odd parts of $x[n]\,u[n]$. Find the simplified expression of

$$\sum_{n=-\infty}^{\infty} x_{ce}[n] \cdot x_{co}^*[n]$$

1. Determine whether the following signals are energy signals, power signals, or neither: (a) $(-0.5)^n u[n]$, (b) $x[n] = u[n]$, (c) $2\,e^{j3n}$, (d) $x[n] = 2^n\,u[n]$. Note that $u[n]$ is the unit step sequence, which is defined as 1 for $n \geq 0$ and 0 for $n < 0$.

Energy signal $\Rightarrow$ E $\rightarrow$ finite , $P_{avg} = 0$

Power signal $\Rightarrow$ $P_{avg} \rightarrow$ finite ($\neq 0$) , $E \rightarrow \infty$

$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$ $\quad P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$

(a) $E = (-0.5)^0 + (-0.5)^2 + \cdots = 4/3$ (Energy signal)

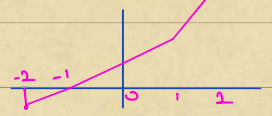
(b) $E = 1+1+\cdots = \infty$ $\quad P = \lim_{N\to\infty} \frac{1}{2N+1}(2N+1) = 1$ (power signal)

(c) $E = 2^2 + 2^2 + \cdots = \infty$ $\quad P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} 4 = 4$ (power signal)

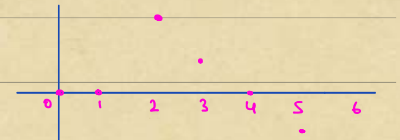
(d) $E = 1 + 2^2 + 2^4 + \cdots = \infty$ $\quad P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} 2^n u[n]$

$= \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} 2^n = \lim_{N\to\infty} \frac{1}{2N+1}(2^N - 1) = \frac{2^N}{2} = \infty$ (neither)

2. Let $x[n] = \{-1,\ 0,\ \underset{\downarrow}{1},\ 2,\ 5\}$ (where the arrow indicates the sample corresponding to the index $n = 0$). Sketch and label carefully each of the following signals: (a) $x[4-n]$, (b) $x[n]\,u[2-n]$, (c) $x[n-1]\,\delta[n-3]$, (d) $x[n^2]$, (e) even part of $x[2n+1]$.



(a)



(b) $x[n]$ (c) $5\delta[n-3]$ (d) $\{2, 1, 1, 2\}$ (e) $\{0, 1, 1.5, 1, 0\}$

4. Let $x[n] \in \mathbb{R}$. Define $x_e[n] = \dfrac{x[n]+x[-n]}{2}$ and $x_o[n] = \dfrac{x[n]-x[-n]}{2}$. Define $E_x = \sum_n |x[n]|^2$. It is claimed that $E_x = E_{x_e} + E_{x_o}$. Prove or disprove the claim.

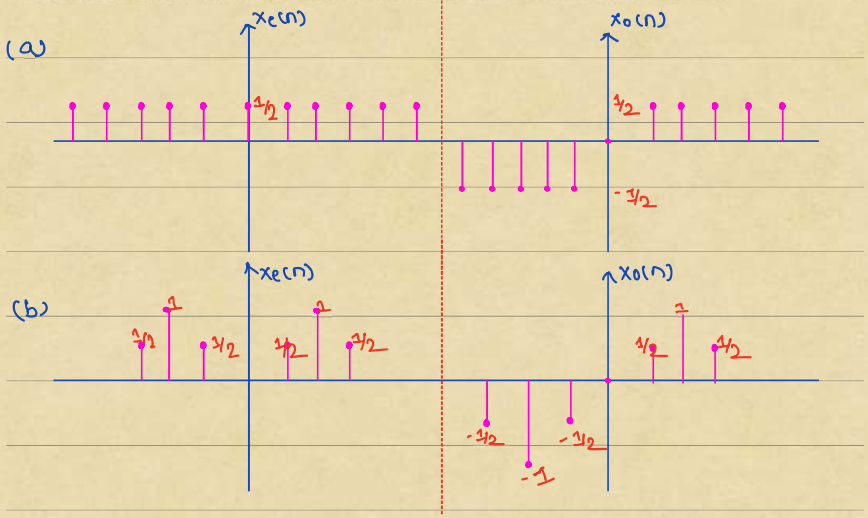
$E_{xe} = \sum_{n=-\infty}^{\infty} |x_e[n]|^2$ $\quad E_{xo} = \sum_{n=-\infty}^{\infty} |x_o[n]|^2$

$E_x = \sum_{n=-\infty}^{\infty} \frac{(x[n])^2 + (x[-n])^2 + 2x[n]x[n]}{4} + \frac{(x[n])^2 + (x[-n])^2 - 2x[n]x[n]}{4}$

$= (x[n])^2$

Prove

5. Sketch the odd and even components of the following signals: (a) $x[n] = 1$ for $0 \le n < 6$, and (b) $x[n] = \{\ldots, 0, \underset{\downarrow}{0}, 1, 2, 1, 0, \ldots\}$.



6. Find the odd and even components of: (a) $u[n]$, (b) $n\,u[n]$, (c) $\sin\left(\frac{n\pi}{4}\right)$, (d) $\cos\left(\frac{n\pi}{4}\right)$, (e) $\sin\left(\frac{n\pi}{4}\right)u[n]$, (f) $\cos\left(\frac{n\pi}{4}\right)u[n]$, (g) $\cos\left(\frac{n\pi}{4}+\frac{\pi}{4}\right)$, (h) $u[n+5]-u[n-5]$

(a) $x_e[n] = \begin{cases} 1/2 & n \neq 0 \\ 1 & n = 1 \end{cases}$ $\quad x_o[n] = \begin{cases} -1/2 & n<0 \\ 0 & n=0 \\ 1/2 & n>0 \end{cases}$

(b) $x_e[n] = \begin{cases} n/2 & n \ge 0 \\ -n/2 & n<0 \end{cases}$ $\quad x_o[n] = n/2$

(c) $x_e[n] = 0$ $\quad x_o[n] = \sin\left(\frac{n\pi}{4}\right)$

(d) $x_e[n] = \cos\left(\frac{n\pi}{4}\right)$ $\quad x_o[n] = 0$

(e) $\frac{\sin\left(\frac{n\pi}{4}\right)u[n] - \sin\left(\frac{n\pi}{4}\right)u[n]}{2} = x_e[n]$

$n>0 \Rightarrow \frac{\sin\left(\frac{n\pi}{4}\right)}{2}$ $\quad n<0 \Rightarrow -\frac{\sin\left(\frac{n\pi}{4}\right)}{2}$ $\Rightarrow x_o[n] = \frac{\sin\left(\frac{n\pi}{4}\right)}{2}$

(f) $x_e[n] = \cos\left(\frac{n\pi}{4}\right)\frac{u[n]+u[-n]}{2} = \frac{1}{2}\cos\left(\frac{n\pi}{4}\right)(1+\delta[n])$

$x_o[n] = \frac{1}{2}\left(\cos\left(\frac{n\pi}{4}\right) - \delta[n]\right)$

(g) $x_e[n] = \frac{\frac{1}{\sqrt{2}}\cos\left(\frac{n\pi}{4}\right) - \frac{1}{\sqrt{2}}\sin\left(\frac{n\pi}{4}\right) + \frac{1}{\sqrt{2}}\cos\left(\frac{n\pi}{4}\right) + \frac{1}{\sqrt{2}}\sin\left(\frac{n\pi}{4}\right)}{2}$

$= 1/\sqrt{2}\cos(n\pi/4)$ $\quad x_o[n] = -1/\sqrt{2}\sin(n\pi/4)$

(h) $x[n] = u[n+5] - u[n-5]$

$\Rightarrow x_e[n] = \frac{u[n+5] - u[n-5] + u[-n+5] - u[-n-5]}{2}$

Find the conjugate symmetric and conjugate anti-symmetric part of $x[n] = e^{jn}$.

$x_{es}[n] = \frac{x[n] + x^*[-n]}{2}$ $\quad x[-n] = e^{-jn} \Rightarrow x^*[-n] = e^{jn}$

$= e^{jn}$; $\quad x_{as} = 0$

8. Let $x_{ce}[n]$ and $x_{co}[n]$ be the conjugate-even and conjugate-odd parts of $x[n]\,u[n]$. Find the simplified expression of

$$z = \sum_{n=-\infty}^{\infty} x_{ce}[n] \cdot x_{co}^*[n]$$

$y[n]$

$x_{ce}[n] = x_{ce}^*[-n]$

$x_{co}[n] = -x_{co}^*[-n]$

$x_{ce}[n]\,x_{co}^*[n]$

$x_{ce}[-n]\,x_{co}^*[-n] = -x_{ce}^*[n]\,x_{co}[n] = -(x_{ce}[n]\,x_{co}^*[n])^*$

$x_{ce}[n]x_{co}^*[n] + x_{ce}[-n]x_{co}^*[-n] = 0$ (always)

10. (a) Simplify the expression for $x(t)\delta(t)$ when $x(t)$ equals: (i) t, and (ii) $\sin t$. Hint: For the first case, consider the limit of $t\,\delta_\Delta(t)$ as $\Delta \longrightarrow 0$, where $\delta_\Delta(t)$ is a symmetric pulse of height $1/\Delta$ and width Δ centred at the origin, i.e.,

$$\delta_\Delta(t) = \frac{1}{\Delta}\left[u\left(t+\frac{\Delta}{2}\right) - u\left(t-\frac{\Delta}{2}\right)\right]$$

(b) What is the value of $\delta(t-\pi)\cos t$?

(a)

(i) $\underset{\Delta \to 0}{\lim}\, t\,\delta(t) = \frac{t(u(t+\Delta/2) - u(t-\Delta/2))}{\Delta} = 0$

(ii) $\sin t\,\delta(t) = \sin(0)\,\delta(t) = 0$

(b) $\delta(t-\pi)\cos t = -\delta(t-\pi)$

9. In the continuous-time framework the complex exponential has the form $x(t) = e^{st}$. Its discrete-time counterpart $x[n]$ can be thought of as the sampled version of $x(t)$, obtained by replacing t by nT, where T is the *sampling period*. Thus, an alternate form for the discrete-time exponential is $x[n] = e^{sTn}$. It is easy to go from one form to the other. Let $x[n] = e^{-n}$. Express it as z_0^n and find z_0. Next, let $x[n] = 2^n$. Express it in the form $e^{c \cdot n}$ and find c. $z_0 = 1/e$ $c = \ln 2$

10. (a) Simplify the expression for $x(t)\delta(t)$ when $x(t)$ equals: (i) t, and (ii) $\sin t$. Hint: For the first case, consider the limit of $t\,\delta_\Delta(t)$ as $\Delta \longrightarrow 0$, where $\delta_\Delta(t)$ is a symmetric pulse of height $1/\Delta$ and width Δ centred at the origin, i.e.,

$$\delta_\Delta(t) = \frac{1}{\Delta}\left[u\left(t + \frac{\Delta}{2}\right) - u\left(t - \frac{\Delta}{2}\right)\right]$$

    (b) What is the value of $\delta(t - \pi)\cos t$?

On a computer, continuous-time signals have to be necessarily represented in sampled form, i.e., replacing t by nT, where T is the sampling interval. A small value of T is required to get good plots and good numerical accuracy; the sampling interval must cover all events of interest. However, too small a value of T will result in a large number of samples, increasing memory and storage requirements.

`MATLAB` and `Scilab` are two very powerful, easy-to-learn, software packages that significantly help in learning concepts taught in this and more advanced courses. Their syntax is very similar. We cannot overstate the importance of learning to use these packages and trying out various computer exercises to fortify the ideas learnt in class. Besides the assigned computer problems, you should explore ideas on your own.

11. **Computer assignment** Given a continuous-time signal specified by

$$x(t) = \begin{cases} 1 - |t| & -1 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$$

    plot the discrete-time sequence that results from uniform sampling of $x(t)$ for the following sampling intervals: (a) 0.25 s, (b) 0.5 s, and (c) 1 s. That is, $x[n] = x(nT)$, where T is the sampling interval. Plot $x(nT)$ versus t and compare that with the plot of $x[n]$ vs n.

12. **Computer assignment** Consider the DT exponential $\mathrm{Re}\{z^n\}$ obtained by sampling the signal $\mathrm{Re}\{e^{st}\}$ with $T = 1$. Using $z = e^{sT}$, plot $\mathrm{Re}\{z^n\}$, as well as $\mathrm{Re}\{e^{st}\}$ for reference, over $-5 \le n \le 5$ for the cases $s = -0.1$ and $s = -0.1 + j2\pi$. Comment on the results.

13. **Computer assignment** Let $x(t) = \cos(2\pi f_0 t)$. Form $x[n] = x(nT)$, where T is called the "sampling period". Let $f_0 = 50$ Hz. Plot the signal $x[n]$ versus nT for a duration of 40ms for various values of T: 125μs, 1 ms, 2 ms, and 10 ms using the default option (which joins the individual points by straight line); repeat to get "stem" plots. Observe what happens when you plot $x[n]$ versus n in all these cases.

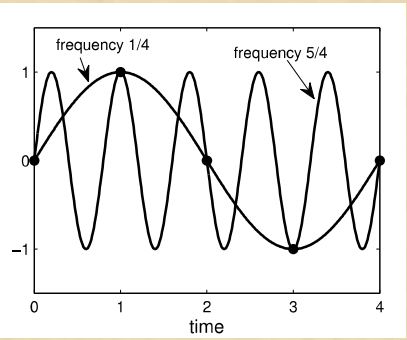
14. **Computer assignment** Plot $(0.9)^n \cos\left(\frac{\pi}{6}n - \frac{\pi}{3}\right)$ and $(1.1)^n \cos\left(\frac{49\pi}{6}n - \frac{\pi}{3}\right)$ for $-10 \le n \le 10$.

15. **Computer assignment** Let $x(t) = A\sin(\omega_0 t + \phi)\exp(-at)$. Let $A = 60$, $\omega_0 = 20\pi$, $\phi = 0$, $a = 6$, and $t = [0, 1]$ (in steps of 1 ms). Plot $x(t)$ versus t.

16. **Computer assignment** Let $x(t) = \sin(2\pi f t)$, where f is some given frequency.
    (a) Choose a convenient time interval and a suitable f and plot the signal.
    (b) Plot the signal after time compression by a factor of 2.
    (c) Plot the signal after time expansion by a factor of 2.
    (d) Plot the signal after time reversal.
    (e) Plot the signal after translating it by a non-zero time.

17. **Computer assignment**
    (a) The figure on the right shows two sinusoids with frequencies $f_1 = 1/4$ and $f_2 = 5/4$. They coincide when $t \in \mathbb{Z}$. Reproduce this plot.

    (b) Let $x(t) = \cos(2\pi f t)$ where $f = \pi$ and $T = 1$ (sampling interval). Is the sampled signal periodic? Plot it over the interval $[0, 50]$.



18. **Computer assignment** The continuous-time delta function and its derivatives can be approximated by discrete-time signals by choosing the sampling time small enough. Choose the sampling interval T very small (i.e., 0.01 or less) and the time axis long enough (e.g., $[-1, 1]$) to encompass all events of interest.
    (a) Define a signal u as the unit step and differentiate it numerically to generate a signal d. Plot d and see how it looks.
    (b) Numerically differentiate d twice more to obtain successively signals d_1 and d_2. Plot d_1 and d_2.
    (c) The signal d is an approximation to the delta function δ, and the signals d_1 and d_2 are approximations to $\delta^{(1)}$ and $\delta^{(2)}$ (note: notation-wise, $x^{(n)}(t)$ denotes the n-th derivative; the zeroth derivative is the signal itself). Define the signal ϕ as

$$\phi(t) = \begin{cases} 1 - 4t^2 & -\frac{1}{2} \le t < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

    and check the validity and accuracy of

$$\int_{-\infty}^{\infty} \delta^{(n)}(t)\,\phi(t)\,dt = (-1)^n \phi^{(n)}(0)$$

    for $n = 0$, 1, and 2. Comment on the results.
    (d) Numerically differentiate the signal ϕ twice over the interval $[-1, 1]$. Explain what you find.

EE2004: Digital Signal Processing

Tutorial 2

(February 5, 2026)

1. The discrete-time sinusoids given below are either in the form $\cos(\omega n+\theta)$ or $\sin(\omega n+\theta)$, where ω can be any finite value. Express each signal in its corresponding alternate form, i.e., either $\cos(\omega_1 n+\theta_1)$ or $\sin(\omega_1 n+\theta_1)$, where $\omega_1 \in [0,\pi]$. (a) $\sin(1.6\pi n+\theta)$, (b) $\cos(2.3\pi n+\theta)$, (c) $\cos(34.6991n+\theta)$, and (d) $\sin(-2.3\pi n+\theta)$.

2. A real-valued sinusoid has frequency $\omega = 2\pi,\ 3\pi,\ 5\pi,\ 3.2\pi,\ 22.1327,\ \pi+2$. To produce an identical signal in each case, what should the corresponding frequency be if it is restricted to the range $[0,\pi]$? In which cases does the phase change sign?

3. What values of θ cause the DT sinusoid $\cos(\omega_0 n+\theta)$ to be a simple shifted version of $\cos(\omega_0 n)$?

4. Find the simplified expression for $x[n]$ where $x[n]=\cos(0.6\pi n+\frac{\pi}{6})+\sqrt{3}\cos(1.4\pi n+\frac{\pi}{3})-2\cos(0.6\pi n-\frac{\pi}{6})$.

5. Express the following exponentials in the form $e^{j(\omega_0 n+\theta)}$ where $\omega_0 \in [-\pi,\pi)$: (a) $e^{j(8.2\pi n+\theta)}$, (b) $e^{j4\pi n}$, (c) $e^{-j1.95n}$, and (d) $e^{-j10.7\pi n}$. Repeat the problem if we require $\omega_0 \in [0,2\pi)$.

6. Consider $x[n]=\cos(\omega_0 n)$ for various values of ω_0 in the set $\{0, 0.1\pi, 0.2\pi, 0.8\pi, 0.9\pi, \pi, 1.1\pi, 1.2\pi, \sqrt{2}\}$. What is the period of $x[\cdot]$ in each case?

7. Determine the period of $x[n]=\sin(0.08\pi n)$. List two other *independent* sinusoids that have the same period as $x[n]$.

8. Consider the harmonically related periodic signals $x_k[n]=\cos(2\pi kn/N)$, where $0 \le k \le N-1$. Since $x_k[n]$ is real-valued, the highest value of k can be limited to the integer that is less than or equal to $N/2$. What is the period of $x_k[n]$ for each k when $N=11$ and $N=12$? Is there any difference in behaviour when compared with its continuous-time counterpart?

9. A system may or may not be (a) memoryless, (b) time-invariant, (c) linear, (d) causal, (e) stable. Determine which of these properties hold and which do not hold for each of the systems whose input-output relationships are given below. Justify your answers. In each example $x[n]$ the system input and $y[n]$ denotes system output. (i) $y[n] = \text{Ev}\{x[n-1]\}$ (where $\text{Ev}\{\cdot\}$ represents the even part),
(ii) $y[n]=\begin{cases} x[n] & n \ge 0 \\ x[n+1] & n \le -1 \end{cases}$, (iii) $y[n]=x[2n]$.

10. Determine whether or not each of the following systems is linear and time-invariant; justify your answer. (a) $y[n]=e^{x[n]}$, (b) $y[n]=x[-n]$, and (c) $y[n]=x[n]+3u[n+1]$.

11. For each system give below, determine which of the following attributes hold: linearity, causality, time-invariance. Wherever necessary assume zero initial conditions. If a property is true, prove it; otherwise, give a counterexample.

1. The discrete-time sinusoids given below are either in the form $\cos(\omega n+\theta)$ or $\sin(\omega n+\theta)$, where ω can be any finite value. Express each signal in its corresponding alternate form, i.e., either $\cos(\omega_1 n+\theta_1)$ or $\sin(\omega_1 n+\theta_1)$, where $\omega_1 \in [0,\pi]$. (a) $\sin(1.6\pi n+\theta)$, (b) $\cos(2.3\pi n+\theta)$, (c) $\cos(34.6991n+\theta)$, and (d) $\sin(-2.3\pi n+\theta)$.

a) $\sin(1.6\pi n+\theta) = \sin(-0.4\pi n+\theta) = \sin(0.4\pi n-\theta+\pi)$

b) $\cos(2.3\pi n+\theta) = \cos(0.3\pi n+\theta)$

c) $\cos(11.04\pi n+\theta) = \cos(-0.96\pi n+\theta) = \cos(0.96\pi n-\theta)$

d) $\sin(-2.3\pi n+\theta) = \sin(-0.3\pi n+\theta) = \sin(0.3\pi n-\theta+\pi)$

2. A real-valued sinusoid has frequency $\omega = 2\pi,\ 3\pi,\ 5\pi,\ 3.2\pi,\ 22.1327,\ \pi+2$. To produce an identical signal in each case, what should the corresponding frequency be if it is restricted to the range $[0,\pi]$? In which cases does the phase change sign?

a) 0 b) π c) π d) 0.8π (phase change)

e) 0.955π (phase change) f) $\pi-2$ (phase changes)

3. What values of θ cause the DT sinusoid $\cos(\omega_0 n+\theta)$ to be a simple shifted version of $\cos(\omega_0 n)$? $\theta = \omega_0 \alpha + 2\pi K \quad K \in \mathbb{R}$

4. Find the simplified expression for $x[n]$ where $x[n]=\cos(0.6\pi n+\frac{\pi}{6})+\sqrt{3}\cos(1.4\pi n+\frac{\pi}{3})-2\cos(0.6\pi n-\frac{\pi}{6})$.

$\cos(0.6\pi n)\frac{\sqrt{3}}{2} - \sin(0.6\pi n)\cdot\frac{1}{2} + \sqrt{3}\cos(1.4\pi n)\frac{1}{2}$

$-\sin(1.4\pi n)\cdot\frac{\sqrt{3}}{2} - 2\cos(0.6\pi n)\frac{\sqrt{3}}{2} - 2\sin(0.6\pi n)\frac{1}{2}$

$= \sqrt{3}\cos(0.6\pi n) + \sin(0.6\pi n) - \sin(0.6\pi n) - \sqrt{3}\cos(0.6\pi n) = 0$

5. Express the following exponentials in the form $e^{j(\omega_0 n+\theta)}$ where $\omega_0 \in [-\pi,\pi)$: (a) $e^{j(8.2\pi n+\theta)}$, (b) $e^{j4\pi n}$, (c) $e^{-j1.95n}$, and (d) $e^{-j10.7\pi n}$. Repeat the problem if we require $\omega_0 \in [0,2\pi)$.

(a) $e^{j(0.2\pi n+\theta)}$ (b) $e^{j(0\pi n)}$ (c) $e^{j(-0.62\pi n)}$ (d) $e^{j(-0.7\pi n)}$

6. Consider $x[n]=\cos(\omega_0 n)$ for various values of ω_0 in the set $\{0, 0.1\pi, 0.2\pi, 0.8\pi, 0.9\pi, \pi, 1.1\pi, 1.2\pi, \sqrt{2}\}$. What is the period of $x[\cdot]$ in each case?

(a) 1 (b) 20 (c) 10 (d) 5 (e) 20 (f) 2 (g) 20 (h) 5 (i) Not periodic

7. Determine the period of $x[n]=\sin(0.08\pi n)$. List two other *independent* sinusoids that have the same period as $x[n]$. period = 25 $\sin(0.16\pi n)$; $\sin(0.24\pi n)$

8. Consider the harmonically related periodic signals $x_k[n]=\cos(2\pi kn/N)$, where $0 \le k \le N-1$. Since $x_k[n]$ is real-valued, the highest value of k can be limited to the integer that is less than or equal to $N/2$. What is the period of $x_k[n]$ for each k when $N=11$ and $N=12$? Is there any difference in behaviour when compared with its continuous-time counterpart? $K=0 \Rightarrow$ period $=1$

$N=11 \Rightarrow$ period $=11$ for $K=1,2,3,4,5$

$N=12 \Rightarrow$ period: $K=1 \Rightarrow 12$ $K=2 \Rightarrow 6$ $K=3 \Rightarrow 4$ $K=4 \Rightarrow 3$ $K=5 \Rightarrow 12$ $K=6 \Rightarrow 2$

9. A system may or may not be (a) memoryless, (b) time-invariant, (c) linear, (d) causal, (e) stable. Determine which of these properties hold and which do not hold for each of the systems whose input-output relationships are given below. Justify your answers. In each example $x[n]$ the system input and $y[n]$ denotes system output. (i) $y[n] = \mathrm{Ev}\{x[n-1]\}$ (where $\mathrm{Ev}\{\cdot\}$ represents the even part),
(ii) $y[n] = \begin{cases} x[n] & n \geq 0 \\ x[n+1] & n \leq -1 \end{cases}$, (iii) $y[n] = x[2n]$.

(i) $y(n) = \frac{x(n-1) + x(-n-1)}{2}$ has memory, linear, non-causal, stable, time variant

(ii) linear, stable, has memory, non-causal, time variant

(iii) linear, has memory, non-causal, time variant, stable

10. Determine whether or not each of the following systems is linear and time-invariant; justify your answer. (a) $y[n] = e^{x[n]}$, (b) $y[n] = x[-n]$, and (c) $y[n] = x[n] + 3u[n+1]$.

(a) non-linear $\alpha x_1 + \beta x_2 \rightarrow \alpha y_1 + \beta y_2$ ✗
time invariant $x(n-n_0) \rightarrow y(n-n_0)$

(b) linear: $\alpha x_1 + \beta x_2 \rightarrow \alpha y_1 + \beta y_2$ ✓
time variant $x(n-n_0) \rightarrow y(n-n_0)$ ✗

(c) nonlinear: $\alpha x_1 + \beta x_2 \rightarrow \alpha y_1 + \beta y_2$ ✗
time variant $x(n-n_0) \rightarrow y(n-n_0)$ ✗

12. Find the simplified expression for $y = x * h$, where $x[n] = a^{-n}u[n]$, $h[n] = -3u[-n-1]$, with $|a| \neq |b|$ and $|a|, |b| > 1$.

$y[n] = \sum_{k=-\infty}^{\infty} a^{-k} u[k] \left(-b^{n-k} u[-n+k-1]\right)$

$= -b^n \sum_{k=-\infty}^{\infty} (ab)^{-k} u[k]\, u[k-(n+1)]$

$n < -1 \Rightarrow -b^n \sum_{k=0}^{\infty} (ab)^{-k} = \frac{-b^n}{1-(ab)^{-1}}$

$n \geq -1 \Rightarrow -b^n \sum_{k=n+1}^{\infty} (ab)^{-k} = \frac{-b^n}{1-(ab)^{-1}} (ab)^{-(n+1)}$

14. Simplify, if possible, the following convolutions: (a) $u[n] * u[n]$, and (b) $u[n] * u[-n]$.

a) $y[n] = \sum_{k=-\infty}^{\infty} u[k]\, u[n-k] = \sum_{k=0}^{n} 1 = (n+1)\, u[n]$

b) $y[n] = \sum_{k=-\infty}^{\infty} u[k]\, u[k-n]$

$n < 0$: $\sum_{k=0}^{\infty} \Rightarrow \infty$
$n > 0$: $\sum_{k=n}^{\infty} \Rightarrow \infty$
} doesn't exist

$y(n) = 10x(n) + 10x(n-2) + 10x(n-6) + \cdots$

(a) $y[n] = y[n-3] + 10x[n]$ linear, causal, T.I

(b) $y[n] = \min_{0 \leq m \leq n} \{x[m]\}$ C, T.V, NL

(c) $y[n] = 2^{|n|}\, x[n]$ L, C, TV

(d) $y[n] = \Re\mathrm{e}\{x[n]\}$ NL $\mathrm{Re}(\alpha x) \neq \alpha \mathrm{Re}(x)$ $\alpha \in C$; C, T.I

(e) $y[n] = \cos(\pi n) + x[n-4]$ NL, C, T.V

(f) $y[n] = \begin{cases} 1 & \text{if } x[n] \geq 0 \\ -1 & \text{otherwise} \end{cases}$ NL, C, T.I

(g) $y[n] = x^*[n]$, NL, C, T.I

(h) $y[n] = -2|x[n]| + \sum_{m=n-2}^{n+2} x[m]$ NL, NC, T.I

12. Find the simplified expression for $y = x * h$, where $x[n] = a^{-n}u[n]$, $h[n] = -b^n u[-n-1]$, with $|a| \neq |b|$ and $|a|, |b| > 1$.

13. Verify the following properties of convolution:

Signal Type	$x[n] = 0$ for	$h[n] = 0$ for	$y[n] = 0$ for	Duration
Right sided	$n < N_1$	$n < N_2$	$n < N_1 + N_2$	
Left sided	$n > N_1$	$n > N_2$	$n > N_1 + N_2$	
Finite Duration	$n < N_1$ $n \geq N_3$ $N_x = N_3 - N_1$	$n < N_2$ $n \geq N_4$ $N_h = N_4 - N_2$	$n < N_1 + N_2$ $n > N_3 + N_4 - 1$	$N_y = N_x + N_h - 1$

14. Simplify, if possible, the following convolutions: (a) $u[n] * u[n]$, and (b) $u[n] * u[-n]$.

15. The *cross-correlation* of two real-valued signals $x[n]$ and $w[n]$ is defined as

$$\phi_{xw}[k] = \sum_{n=-\infty}^{\infty} x[n+k]\, w[n]$$

The special case when $x[n] = w[n]$ is called the *autocorrelation* function.

(a) Express $\phi_{xw}[k]$ as a convolution.

(b) Find the relationship between $\phi_{xw}[k]$ and $\phi_{wx}[k]$ and also between $\phi_{xx}[k]$ and $\phi_{xx}[-k]$.

(c) Let $x[n] = u[n] - u[n-N+1]$. Plot (a) $y[n] = x[n] * x[n]$, and (b) $\phi_{xx}[k]$.

16. (a) Working in the time-domain and using a step-by-step procedure, assuming $x[n] = \delta[n]$, find the output the system described by the following difference equation: $y[n] - 0.8\, y[n-1] = x[n]$. Assume $y[-1] = 0$.

(b) Repeat when the input is $0.5\, u[n]$; as before, assume $y[n] = 0$ for $n < 0$. Compare this answer with the result of $0.5\, u[n] * 0.8^n\, u[n]$? Can you relate this to the system output obtained in part (a) ?

17. $y[n] = x[n] * h[n]$. Let $h[n] = 0$ outside $0 \leq n \leq N-1$, which reduces the convolution summation to a finite one. For each n, the output $y[n]$ can be thought of as an *inner product* of two N-dimensional vectors, i.e., $\mathbf{a}^T\mathbf{b}$. Let $\mathbf{y}$ represent in vector form the output from 0 to $N-1$. The convolution can be expressed as $\mathbf{y} = \mathbf{X}\mathbf{h}$. What are the entries of $\mathbf{X}$ and $\mathbf{h}$? What can you say about the diagonal entries of $\mathbf{X}$?

15. The *cross-correlation* of two real-valued signals $x[n]$ and $w[n]$ is defined as

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(c) Let $x[n] = u[n] - u[n-N+1]$. Plot (a) $y[n] = x[n] * x[n]$, and (b) $\phi_{xx}[k]$.

(a) $x_{xw}[k] = \sum_{n=-\infty}^{\infty} x[n+k]w[n] = x[k] * w[-k]$

(b) $x_{wx}[k] = \sum_{n=-\infty}^{\infty} w[n+k]x[n] = w[k] * x[-k]$

$x_{xw}[k] = x_{wx}[-k]$

(c) $x[n] = u[n] - u[n-(N-1)]$

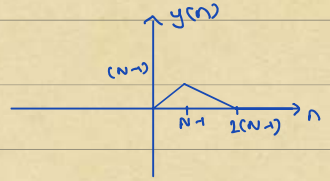
$x[n] * x[n] = \sum_{K=-\infty}^{\infty} \big(u[k] - u[k-(N-1)]\big)\big(u[n-k] - u[n-k-(N-1)]\big)$

$N-1 < n-(N-1) \Rightarrow 0$

$n > N-1 > n-(N-1) \Rightarrow 2(N-1)-n$

$N-1 > n \Rightarrow n-1$

$$y(n) = \begin{cases} n-1 & n < N-1 \\ 2(N-1)-(n-1) & n \geq N-1 \;;\; n < 2(N-1) \\ 0 & n > 2(N-1) \end{cases}$$



$\phi_{xx}[k] = \sum_{n=-\infty}^{\infty} x[n+k]\,x[n]$

$= \sum_{n=-\infty}^{\infty} \big(u[n+k] - u[n+k-(N-1)]\big)\big(u[n] - u[n-(N-1)]\big)$

$n \in [-k, (N-1)-k] \quad n \in [0, N-1]$

$(N-1)-k < 0 \Rightarrow k > N-1 \Rightarrow 0$

$-k > N-1 \Rightarrow k < -(N-1) \Rightarrow 0$

$(N-1)-k > 0 \;;\; -k < 0 \Rightarrow k \in [0, N-1) \Rightarrow (N-k)$

$\Rightarrow \quad N-|k| \quad |k| \leq N-1$

$0 \quad$ else

16. (a) Working in the time-domain and using a step-by-step procedure, assuming $x[n] = \delta[n]$, find the output the system described by the following difference equation: $y[n] - 0.8\,y[n-1] = x[n]$. Assume $y[-1] = 0$.

(b) Repeat when the input is $0.5\,u[n]$; as before, assume $y[n] = 0$ for $n < 0$. Compare this answer with the result of $0.5\,u[n] * 0.8^n\,u[n]$? Can you relate this to the system output obtained in part (a) ?

(a) $n=0 \Rightarrow y[0] = x[0] = 1$

$n=1 \Rightarrow y[1] - 0.8 = \delta[1] \Rightarrow y[1] = 0.8$

$n=2 \Rightarrow y[2] - 0.8^2 = 0 \Rightarrow y[2] = 0.8^2 \quad \Rightarrow y[n] = 0.8^n u[n]$

(b) $h[n] = 0.8^n u[n] \quad x[n] = 0.5\,u[n]$

$y[n] = x[n] * h[n] = 0.5u[n] * 0.8^n u[n]$

$= \sum_{k=-\infty}^{\infty} 0.5\,u[k] \cdot 0.8^{n-k} u[n-k]$

$= 0.8^n \times 0.5 \sum_{k=0}^{n} 0.8^{-k} = 0.5 \times 0.8^n \left(\frac{1-0.8^{n+1}}{0.2}\right)$

$= 0.8^n (1 - 0.8^{n+1}) \times 1.5$

17. $y[n] = x[n] * h[n]$. Let $h[n] = 0$ outside $0 \leq n \leq N-1$, which reduces the convolution summation to a finite one. For each n, the output $y[n]$ can be thought of as an *inner product* of two N-dimensional vectors, i.e., $\mathbf{a}^T\mathbf{b}$. Let $\mathbf{y}$ represent in vector form the output from 0 to $N-1$. The convolution can be expressed as $\mathbf{y} = \mathbf{Xh}$. What are the entries of $\mathbf{X}$ and $\mathbf{h}$? What can you say about the diagonal entries of $\mathbf{X}$?

$y[n] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k] = \sum_{k=0}^{N-1} h[k]\,x[n-k]$

$= x[n]h[0] + x[n-1]h[1] + \cdots + x[n-(N-1)]h[N-1]$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ \vdots \\ y[N-1] \end{bmatrix} = \begin{bmatrix} x[0] & x[-1] & \cdots & x[-(N-1)] \\ x[1] & x[0] & \cdots & x[-(N-2)] \\ \vdots & \vdots & & \vdots \\ x[N-1] & x[N-2] & & x[0] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[N-1] \end{bmatrix}$$

18. Suppose $y = x * h$. $\{x\} = \{\underset{\downarrow}{2},2,3,3,4\}$ and $\{y\} = \{\underset{\downarrow}{2},4,9,10,13,10,8\}$. Find $\{h\}$.

$h = \{h_0, h_1, h_2\} \qquad y[n] = \sum x[k]\,h[n-k]$

$n=0 \Rightarrow 2 = \sum x[k]h[-k] = 2h_0 \Rightarrow h_0 = 1$

$n=1 \Rightarrow 4 = \sum x[k]h[1-k] = 2h_1 + 2\cdot 1 \Rightarrow h_1 = 1$

$n=2 \Rightarrow 9 = 2h_2 + 2\cdot 1 + 3\cdot 1 \Rightarrow h_2 = 2$

$\{1, 1, 2\}$

18. Suppose $y = x * h$. $\{x\} = \{2,2,3,3,4\}$ and $\{y\} = \{2,4,9,10,13,10,8\}$. Find $\{h\}$.

19. If you multiply two polynomials, the coefficients of the result can be obtained by considering each polynomial's coefficients as a sequence and convolving the two. Consider $\{1,3,1\} * \{1,2,2\}$. Use this result to compute 131×122.

20. Consider an LTI system with $h[n] = 2^{-n}\,u[n]$. Find the output $y[n]$ if the input $x[n] = u[n] - u[n-10]$.

21. **Computer assignment** Consider the convolution of $\{1,2,3,4\}$ and $\{3,2,1\}$. Determine the 6×4 matrix $\mathbf{H}$ such that $\mathbf{y} = \mathbf{H}\mathbf{x}$. Since convolution is commutative, obtain $\mathbf{y}$ by using the equation $\mathbf{X}\mathbf{h}$. What are the entries of $\mathbf{X}$? A convolution matrix can be quickly constructed using the `toeplitz` command; see also `convmtx`.

22. **Computer assignment** Let $\mathbf{y} = \mathbf{X}\mathbf{h}$, where the matrices are as defined in the previous problem. Consider the sequences given in Problem 18. Let $\mathbf{h} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$. Does this result agree with what you obtained earlier? Look up material on the least-squares solution of $\mathbf{A}\mathbf{x} = \mathbf{y}$. Can you obtain $\mathbf{h}$ by solving $\mathbf{y} = \mathbf{X}\mathbf{h}$ recursively?

23. **Computer Experiment** To generate a sequence with frequency components 0.1 and 0.2, first define `n=(0:99)';` to get a column vector of length 100. The sequence `x` can then be generated using the command `x = cos(2*pi*0.1*n+pi/3) + sin(2*pi*0.2*n + pi/8);`. The higher frequency component can be eliminated by convolving `x` with `h`, where `h = 0.2*ones(5,1);`. Plot `y` where `y = conv(x,h)`. Verify that the second component is practically absent in `y`. Give a theoretical explanation as to why the convolution with an all ones sequence of length 5 has eliminated the sinusoid with frequency 0.2.

24. **Computer Experiment** Form the sequence $h[n] = a^n\,u[n]$ for $a = 0.9$. Let $x[n] = \sin(2\pi n/16)$ for $n = 0,1,...,999$. Compute the convolution of these two sequences and plot the first hundred samples. The built-in commands for convolution are `convol` (Scilab) and `conv` (Matlab).

    Calculate the theoretical convolution output when $h[n] = (0.9)^n\,u[n]$ is convolved with $x[n] = \sin(2\pi n/16)\ \forall n$. The result can be decomposed into two terms: $\alpha\,0.9^n\,u[n] + \beta\,\sin(2\pi n/16 + \gamma)$. Calculate the constants α, β, and γ and compare the theoretical results with that obtained from Matlab/Scilab.

25. **Computer assignment** Consider the convolution of $x[n]$ and $y[n]$, where $y[n] = x[-n]$. The result is called the *autocorrelation* of $x[n]$. Let `x[n] = rand(100,1)`. Find the autocorrelation of $x[n]$. Normalize the peak value to unity. At what time index does it occur? Replace `rand` by `randn` and repeat. Do you see any qualitative change in the result? Study the function `xcorr`. If the samples of a sequence are uncorrelated, the resulting autocorrelation sequence will be impulsive (or approximately so). For example, consider N successive digits of π. Replace an even digit by 1 and and odd digit by 0. Find the autocorrelation of the resulting sequence. The first ten thousand digits of π can be found at `http://www.joyofpi.com/pi.html`. Repeat the experiment by replacing the even digits by 1 and the odd-digits by -1. What is the main difference that you observe in this case? ◇

19. If you multiply two polynomials, the coefficients of the result can be obtained by considering each polynomial's coefficients as a sequence and convolving the two. Consider $\{1,3,1\} * \{1,2,2\}$. Use this result to compute 131×122.

$\{1,3,1\}$ $\{1,2,2\}$ $x[k]\,h[n-k]$

$1\times2 = 2$
$1\times2+3\times1 = 5$
$1\times2+3\times2+1\times1 = 9$ $\{2,5,9,8,2\}$
$3\times2+1\times2 = 8$
$1\times2 = 2$

20. Consider an LTI system with $h[n] = 2^{-n}\,u[n]$. Find the output $y[n]$ if the input $x[n] = u[n] - u[n-10]$.

$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} (u[k] - u[k-10])\cdot 2^{-(n-k)}\,u[n-k]$

$= 2^{n}\cdot\sum_{k=0}^{9} 2^{k}\,u[n-k]$

$n<0 \Rightarrow 0$
$n\in[0,9] \Rightarrow 2^{n}\sum_{k=0}^{n} 2^{k} = 2^{n}\cdot(2^{n+1}-1) = 1-2^{-n}$
$n>10 \Rightarrow 2^{n}\cdot\sum_{k=0}^{10} 2^{k} = 2^{n}(2^{10}-1)$

$a^n u[n] \rightarrow \frac{1}{1-az^{-1}} \quad |z|>|a|$

$-a^n u[-n-1] \rightarrow \frac{1}{1-az^{-1}} \quad |z|<|a|$

EE2004: Digital Signal Processing

Tutorial 3

(February 12, 2026)

1. Find the z-transform of the following sequences and state clearly the RoC in each case: (a) $(0.8)^n\,u[n] + 2^n\,u[-n-1]$, (b) $2^n\,u[n] - 3^n\,u[-n-1]$, (c) $(0.8)^n\,u[n] + (0.9)^n\,u[-n-1]$, (d) $[(0.8)^n + 3\,(0.4)^n]\,u[-n-1]$, (e) $[(0.8)^n + 3\,(0.4)^n]\,u[n]$, and (f) $(0.5)^{|n|}$.

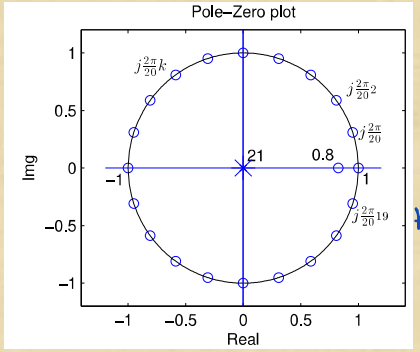
2. Let $x[n] = e^{j\omega_0 n}\,u[n]$. Find $X(z)$ and its corresponding RoC.

3. Let $x[n] = \cos(\omega_0 n)\,u[n]$. Find $X(z)$ and its corresponding RoC. Repeat for $x[n] = \sin(\omega_0 n)\,u[n]$. In each case, plot the pole-zero plot of $X(z)$.

4. Let $H(z) = \dfrac{(e^{-2}-2)z}{(z-e^{-2})(z-2)}$. List all possible RoCs corresponding to the given $H(z)$. Also state the nature of the corresponding time-domain sequence for each case, i.e., left-sided, right-sided, or two-sided.

5. Find $X(z)$ when $x[n]$ equals (a) $\delta[n]$, (b) $\delta[n-n_0]$ $(n_0 > 0)$, and (c) $\delta[n+n_0]$ $(n_0 > 0)$. What is the RoC in each case?

6. Find $X(z)$ when $x[n] = u[n]$ (the unit step sequence). What is its RoC?

7. 

Consider the pole-zero plot shown on the left. Find the corresponding $H(z)$ whose pole-zero plot is consistent with the given information.

$K\,\frac{(z^{20}-1)(z-0.8)}{z^{21}}$

8. Let $x[n] = r^n\,\dfrac{\sin[(n+1)\theta]}{\sin\theta}\,u[n]$. Find $X(z)$ and indicate the RoC. You should not assume the z-transform of either $\cos(\omega_0 n)\,u[n]$ or $\sin(\omega_0 n)\,u[n]$ while deriving the final expression.

9. **Computer assignment** Create the signal $x[n] = \cos\left(\dfrac{n\pi}{5}\right)$ for $0 \le n \le 19$. Delay this signal by 2.5 samples as follows: (i) multiply the signal's DTFT (implemented using Matlab's `fft` command) by $\exp(-j\,2.5\omega)$ (after discretizing the frequency variable) over the the interval $[0, \pi]$; (ii) for the samples in the range $(\pi, 2\pi)$, fill them using the time-reversed, complex-conjugated samples taken from the interval $[0, \pi]$; (iii) find

1. Find the z-transform of the following sequences and state clearly the RoC in each case: (a) $(0.8)^n\,u[n] + 2^n\,u[-n-1]$, (b) $2^n\,u[n] - 3^n\,u[-n-1]$, (c) $(0.8)^n\,u[n] + (0.9)^n\,u[-n-1]$, (d) $[(0.8)^n + 3\,(0.4)^n]\,u[-n-1]$, (e) $[(0.8)^n + 3\,(0.4)^n]\,u[n]$, and (f) $(0.5)^{|n|}$.

$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

(a) $X(z) = \sum_{n=-\infty}^{\infty} \left((0.8)^n u[n] + 2^n u[-n-1] \right) z^{-n}$

$= \sum_{n=-\infty}^{\infty} (0.8)^n u[n]\, z^{-n} + \sum_{n=-\infty}^{\infty} 2^n u[-n-1] z^{-n}$

$= \sum_{n=0}^{\infty} (0.8/z)^n + \sum_{n=-\infty}^{-1} (z/2)^n$

$= \frac{1}{1-0.8/z} + \sum_{n=1}^{\infty} (z/2)^n \qquad \left|\frac{0.8}{z}\right| < 1 \quad |z| > 0.8$

$= \frac{z}{z-0.8} + \frac{z/2}{1-z/2} \qquad \left|\frac{z}{2}\right| < 1 \quad |z| < 2$

$= \frac{z}{z-0.8} + \frac{z}{2-z} \qquad \text{ROC: } 0.8 < |z| < 2$

(b) $X(z) = \sum_{n=-\infty}^{\infty} \left(2^n\,u[n] - 3^n\,u[-n-1]\right) z^{-n}$

$= \sum_{n=0}^{\infty} (2/z)^n - \sum_{n=-\infty}^{-1} (3/z)^n$

$= \frac{1}{1-2/z} - \sum_{n=1}^{\infty} (z/3)^n \qquad |z|>2,\ |z|<3$

$= \frac{z}{z-2} - \frac{z/3}{1-z/3} = \frac{z}{z-2} + \frac{z}{z-3} \qquad 2 < |z| < 3$

(c) $X(z) = \sum_{n=0}^{\infty} (0.8/z)^n + \sum_{n=1}^{\infty} (z/0.9)^n$

$= \frac{1}{1-0.8/z} + \frac{z/0.9}{1-z/0.9} = \frac{z}{z-0.8} + \frac{z}{0.9-z} \qquad 0.8 < |z| < 0.9$

(d) $\frac{-z}{z-0.8} - \frac{3z}{z-0.4} \qquad |z| < 0.4$

(e) $\frac{z}{z-0.8} + \frac{3z}{z-0.4} \qquad |z| > 0.8$

(f) $X(z) = \sum_{n=-\infty}^{\infty} 0.5^{|n|} z^{-n} = \sum_{n=-\infty}^{-1} (0.5z)^{-n} + \sum_{n=0}^{\infty} (0.5/z)^n$

$= \frac{0.5z}{1-0.5z} + \frac{1}{1-0.5/z} \qquad 2 > |z| > 0.5$

2. Let $x[n] = e^{j\omega_0 n}\, u[n]$. Find $X(z)$ and its corresponding RoC.

$X(z) = \sum_{n=-\infty}^{\infty} e^{j\omega_0 n} u[n] z^{-n} = \sum_{n=0}^{\infty} \left[e^{j\omega_0}/z\right]^n = \frac{1}{1 - e^{j\omega_0} z^{-1}}$ $\quad |z| > 1$

3. Let $x[n] = \cos(\omega_0 n)\, u[n]$. Find $X(z)$ and its corresponding RoC. Repeat for $x[n] = \sin(\omega_0 n)\, u[n]$. In each case, plot the pole-zero plot of $X(z)$.

$\cos(\omega_0 n) = \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}$ $\quad \sin(\omega_0 n) = \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}$

4. Let $H(z) = \frac{(e^{-2} - 2)z}{(z - e^{-2})(z - 2)}$. List all possible RoCs corresponding to the given $H(z)$. Also state the nature of the corresponding time-domain sequence for each case, i.e., left-sided, right-sided, or two-sided.

poles are: $2, e^{-2}$

$|z| < e^{-2} \rightarrow$ left side

$e^{-2} < |z| < 2 \rightarrow$ two-sided

$|z| > 2 \rightarrow$ right sided

5. Find $X(z)$ when $x[n]$ equals (a) $\delta[n]$, (b) $\delta[n - n_0]$ $(n_0 > 0)$, and (c) $\delta[n + n_0]$ $(n_0 > 0)$. What is the RoC in each case?

(a) 1; finite z-plane

(b) z^{n_0}; finite z-plane except $z = 0$

(c) z^{n_0}; finite z-plane except $z = \infty$

6. Find $X(z)$ when $x[n] = u[n]$ (the unit step sequence). What is its RoC? $\frac{1}{1 - z^{-1}}$ $\quad |z| > 1$

8. Let $x[n] = r^n \frac{\sin[(n+1)\theta]}{\sin\theta}\, u[n]$. Find $X(z)$ and indicate the RoC. You should not assume the z-transform of either $\cos(\omega_0 n)\, u[n]$ or $\sin(\omega_0 n)\, u[n]$ while deriving the final expression.

$$X(z) = \sum_{n=0}^{\infty} r^n \frac{\sin[(n+1)\theta]}{\sin\theta} z^{-n}$$

$$= \frac{1}{\sin\theta} \sum_{n=0}^{\infty} \frac{r^n z^{-n} \left(e^{j(n+1)\theta} - e^{-j(n+1)\theta}\right)}{2j}$$

$$= \frac{e^{j\theta}}{2j\sin\theta} \sum_{n=0}^{\infty} \left(re^{j\theta}/z\right)^n - \frac{e^{-j\theta}}{2j\sin\theta} \sum_{n=0}^{\infty} \left(r/ze^{j\theta}\right)^n$$

$$= \frac{e^{j\theta}}{2j\sin\theta} \frac{1}{1 - re^{j\theta} z^{-1}} - \frac{e^{-j\theta}}{2j\sin\theta} \frac{1}{1 - re^{-j\theta} z^{-1}} \quad |z| > r$$

$$= \frac{1}{2j\sin\theta} \left\{ \frac{1}{e^{j\theta} - rz^{-1}} - \frac{1}{e^{-j\theta} - rz^{-1}} \right\}$$

$$= \frac{1}{2j\sin\theta} \frac{e^{j\theta} - e^{-j\theta}}{r^2 z^{-2} - 2rz^{-1}\cos\theta + 1} = \frac{1}{r^2 z^{-2} - 2r\cos\theta z^{-1} + 1} \quad |z| > r$$

the inverse transform to obtain a sequence that is 20 samples in length (it must be real-valued, and any imaginary part present must be very small and should be zeroed out). Plot this alongside of $y[n] = \cos\left(\frac{n\pi}{5} - \frac{\pi}{2}\right)$. Also compare this with `delayseq(x, 2.5)`.

10. **Computer assignment** Load the speech waveform contained in the file `f0125s.wav` (look up Matlab's `audioread` command). Add a delayed version of the speech to itself. Consider delays in the range 50–100 ms. Note the difference in sound quality. Increase the delay and observe what happens for values much greater than 100 ms.

11. **Computer assignment** Load the speech file `f0103s.wav`. Pass it through a block that introduces a *time-varying* delay. That is, this block produces the output is $x[n - d_n]$. Choose `d = 20 + 20*sin(2*pi*5*n/Fs)`. Implement this block using Matlab's `delayseq` command, which has the capability to introduce fractional delays. A *vibrato* effect is imparted to the output. Experiment with variable delay parameters to see how they influence the vibrato effect.

12. **Computer assignment** Load the speech file `f0126s.wav`. Consider the following two different variable delay values, i.e., `d1 = 200 + 32*sin(2*pi*2*n/Fs)` and `d2 = 200 + 50*sin(2*pi*3*n/Fs)`. Pass the original signal through each of these two variable delay blocks. Listen to each output and compare it with the original. In particular, are you able to note the changes in the words "along" and "one mile lane"?; are there any other noticeable differences? Next, add the two outputs to the original and listen to combined signal. This produces a *chorus* effect. Vary the parameters of the delay blocks and observe how they change the sound quality.

EEE2004: Digital Signal Processing

Tutorial 4

(February 19, 2026)

1. Determine the RoC of the z-transform of the following sequences without explicitly finding $X(z)$; in each case specify whether or not the unit circle is part of the RoC.
   (a) $x[n] = \left[\left(\frac{1}{2}\right)^n - \left(-\frac{4}{3}\right)^n\right] u[n-10]$ $|z| > 4/3$; no
   (b) $x[n] = \left(-\frac{10}{9}\right)^n u[-n+4]$ $|z| < 10/9$; yes
   (c) $x[n] = \left(\frac{e}{\pi}\right)^n u[n+4]$ $|z| > e/\pi$; yes
   (d) $x[n] = \left(\frac{e}{-\pi}\right)^n u[n] + (2+3j)^{n-2} u[-n-1]$ $e/\pi < |z| < \sqrt{13}$; yes
   (e) $x[n] = \sin(n)\, u[n-1]$ $|z| > 1$; no
   (f) $x[n] = \dfrac{\sin(n)}{2^n} u[n-1]$ $|z| > 1/2$; yes

2. Consider the finite-duration sequence $h[n]$ defined in the range $0 \le n \le N-1$. Let $g[n] = h^*[N-1-n]$.
   (a) Express $G(z)$ in terms of $H(z)$.
   (b) Now let $g[n] = h[n]$, i.e., $h^*[N-1-n] = h[n]$. If z_0 is a zero of $H(z)$, can you find another zero related to z_0?
   (c) Suppose now $h[n]$ is real-valued and z_0 is a non-trivial zero that is not located on the unit circle. How many other zeros of $H(z)$ can you determine from the knowledge of z_0?

3. State which of the following are true/false. Justify your answer with proof if true, or with a counterexample if false:
   (a) For any sequence, whose z-transform exists and well-defined, its DTFT also exists and can be computed by evaluating the z-transform along the unit circle. false
   (b) If two sequences $x_1[n]$ and $x_2[n]$ have well-defined z-transforms with RoCs R_1 and R_2 respectively, then any linear combination of the two sequences will have well-defined z-transform with RoC $R \subseteq R_1 \cap R_2$. false $a^n u[n]$; $-a^n u[n]$
   (c) Suppose $X(z)$ is a well-defined z-transform of $x[n]$ with poles at $z = z_k$, where $k \in \{1, 2, \cdots, N\}$ and z_k's are non-zero and finite values. If $X(z)$ the number of possible RoCs is exactly two, then $z_1 = z_2 = \cdots = z_N$. false (magnitudes should be equal)

4. A causal LTI discrete-time system is described by the input/output equation $y[n] = 0.2\,y[n-1] + 0.08\,y[n-2] + 2\,x[n]$. Find the system transfer function $H(z)$.

5. Consider the transform pair $x[n] \xleftrightarrow{Z} X(z)$ with RoC $|z| > r > 1$. For what range of values $a \in \mathbb{C}$ is the sequence $y[n] = a^n\, x[n]$ absolutely summable? Justify your answer.

6. Let $x[n] = 0.5^n u[n]$ and $y[n] = x[n] * x[n]$, where '$*$' denotes convolution. Evaluate $\sum_{n=-\infty}^{\infty} y[n]$. $Y(1) = X(1) \times X(1) = \left(\frac{1}{1-0.5}\right)^2 = 4$

1. Determine the RoC of the z-transform of the following sequences without explicitly finding $X(z)$; in each case specify whether or not the unit circle is part of the RoC.
   (a) $x[n] = \left[\left(\frac{1}{2}\right)^n - \left(-\frac{4}{3}\right)^n\right] u[n-10]$
   (b) $x[n] = \left(-\frac{10}{9}\right)^n u[-n+4]$
   (c) $x[n] = \left(\frac{e}{\pi}\right)^n u[n+4]$
   (d) $x[n] = \left(\frac{e}{-\pi}\right)^n u[n] + (2+3j)^{n-2} u[-n-1]$
   (e) $x[n] = \sin(n)\, u[n-1]$
   (f) $x[n] = \dfrac{\sin(n)}{2^n} u[n-1]$

(a) $\sum_{n=10}^{\infty} (1/2)^n z^{-n} - \sum_{n=10}^{\infty} (-4/3)^n z^{-n}$

$= \dfrac{(1/2)^{10} z^{-10}}{1-(1/2)z^{-1}} - \dfrac{(-4/3)^{10} z^{-10}}{1-(-4/3)z^{-1}}$ $|z| > \frac{4}{3}$

2. Consider the finite-duration sequence $h[n]$ defined in the range $0 \le n \le N-1$. Let $g[n] = h^*[N-1-n]$.
   (a) Express $G(z)$ in terms of $H(z)$.
   (b) Now let $g[n] = h[n]$, i.e., $h^*[N-1-n] = h[n]$. If z_0 is a zero of $H(z)$, can you find another zero related to z_0?
   (c) Suppose now $h[n]$ is real-valued and z_0 is a non-trivial zero that is not located on the unit circle. How many other zeros of $H(z)$ can you determine from the knowledge of z_0?

$G(z) = \sum_{n=-\infty} g(n) z^{-n} = \sum_{n=0}^{N-1} h^*(N-1-n) z^{-n} = \sum_{n=0}^{N-1} h^*(n) z^{-n-(N-1)}$

$= z^{-(N-1)} \sum_{n=0}^{N-1} h^*(n) z^{n} = z^{-(N-1)} H^*(1/z^*)$

$H(z) \to h(z)$

$H^*(z^*) \to h^*(z)$

4. A causal LTI discrete-time system is described by the input/output equation $y[n] = 0.2\,y[n-1] + 0.08\,y[n-2] + 2\,x[n]$. Find the system transfer function $H(z)$.

$y(n) = 0.2y(n-1) + 0.08y(n-2) + 2x(n)$

$y(n) z^{-n} = 0.2y(n-1)z^{-n} + 0.08\,y(n-2)z^{-n} + 2x(n) z^{-n}$

$\sum_{n=-\infty} y(n) z^{-n} = 0.2\, z^{-1} \sum_{n=-\infty} y(n-1) z^{-(n-1)}$
$+ 0.08\, z^{-2} \sum_{n=-\infty} y(n-2) z^{-(n-2)} + 2 \sum_{n=-\infty} x(n) z^{-n}$

$\Rightarrow Y(z) = 0.2\, z^{-1} Y(z) + 0.08 z^{-2} Y(z) + 2X(z)$

$\Rightarrow Y(z)\left[1 - \frac{0.2}{z} - \frac{0.08}{z^2}\right] = 2X(z)$

$\Rightarrow \dfrac{Y(z)}{X(z)} = \dfrac{2z^2}{z^2 - 0.2z - 0.08} = H(z)$

5. Consider the transform pair $x[n] \xleftrightarrow{Z} X(z)$ with RoC $|z| > r > 1$. For what range of values $a \in \mathbb{C}$ is the sequence $y[n] = a^n\, x[n]$ absolutely summable? Justify your answer.

$Y(z) = \sum_{n=-\infty} a^n x(n) z^{-n} = X(z/a)$ $|z/a| > r$

$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$ $Y(1) = \sum_{n=-\infty}^{\infty} a^n x(n) = X(1/a)$

$\Rightarrow 1/|a| > \gamma \Rightarrow |a| < \frac{1}{\gamma}$

$x[n] = n u[n]$

7. Given $X(z) = \frac{z^{-1}}{(1-z^{-1})^2}$, determine the transform of the following signals using z-transform properties and specify the RoC (a) $x[n-3]$, (b) $x[n] \cdot \delta[n-3]$, (c) $x[n] - x[n-1]$, (d) $x[n] \cdot (\delta[n] - \delta[n-1])$, (e) $5\,x[n-1] + 4\left(-\frac{1}{3}\right)^n u[n]$.

(a) $z^{-3}X(z)$ (b) $3z^{-3}$ (c) $X(z) - z^{-1}X(z)$ (d) $-z^{-1}$ (e) $5z^{-1}X(z) + \frac{4}{1+\frac{1}{3}z^{-1}}$

8. Using the differentiation propery, find the sequences whose z-transforms are:
   (a) $X_1(z) = \log(1 - az^{-1})$ for $|z| > |a|$
   (b) $X_2(z) = \log\left(\frac{a}{a - z^{-1}}\right)$ for $|z| > \frac{1}{|a|}$
   (c) $X_3(z) = \log\left(1 - a^{-1}z\right)$ for $|z| < |a|$

$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$\frac{d}{dz}X(z) = \sum_{n=-\infty}^{\infty} x(n)(-n) z^{-(n+1)}$

$-z\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} n\,x(n) z^{-n} = n \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$-z\frac{dX(z)}{dz} \longrightarrow n\,x(n)$

a) $\frac{d}{dz}\log(1 - az^{-1}) = \frac{1}{1 - az^{-1}} \cdot az^{-2}$ $\quad x_1 \Rightarrow \log$

$z\frac{dx_1}{dz} = \frac{a z^{-1}}{1 - az^{-1}}$

$n x_1(n) = -a^n u[n-1] \Rightarrow x_1(n) = -\frac{a^n}{n} u[n-1]$

b) $\frac{d}{dz}\log\left(\frac{a}{a - z^{-1}}\right) = \frac{a - z^{-1}}{a} \cdot \frac{a}{(a - z^{-1})^2} \cdot z^{-2} = \frac{z^{-2}}{(a - z^{-1})}$

$-z\frac{dx_2}{dz} = \frac{-a z^{-1}}{1 - a z^{-1}} \Rightarrow n x_2(n) = -a^{(n)} u[n-1]$

$\Rightarrow x_2(n) = -\frac{a^n}{n} u[n-1]$

c) $\frac{d}{dz}\log(1 - a^{-1}z) = \frac{-a^{-1}}{1 - a^{-1}z}$

$-z\frac{dx_3}{dz} = \frac{-a^{-1}z}{1 - a^{-1}z} = \frac{1}{1 - az^{-1}} \Rightarrow n x_3(n) = a^n u[-n-1]$

$\Rightarrow x_3(n) = \frac{a^n}{n} u[-n-1]$

7. Given $X(z) = \frac{z^{-1}}{(1-z^{-1})^2}$, determine the transform of the following signals using z-transform properties and specify the RoC (a) $x[n-3]$, (b) $x[n] \cdot \delta[n-3]$, (c) $x[n] - x[n-1]$, (d) $x[n] \cdot (\delta[n] - \delta[n-1])$, (e) $5\,x[n-1] + 4\left(-\frac{1}{3}\right)^n u[n]$.

8. Using the differentiation propery, find the sequences whose z-transforms are:
   (a) $X_1(z) = \log(1 - az^{-1})$ for $|z| > |a|$
   (b) $X_2(z) = \log\left(\frac{a}{a - z^{-1}}\right)$ for $|z| > \frac{1}{|a|}$
   (c) $X_3(z) = \log\left(1 - a^{-1}z\right)$ for $|z| < |a|$

9. Use z-transform properties to find the convolution result of $x[n]$ and $h[n]$, where $x[n] = \delta[n] - 2\,\delta[n-2]$ and $h[n] = 2\,\delta[n] - 2\,\delta[n-1] + 3\,\delta[n-2] + \delta[n-3]$.

10. Let $x[n] = (0.3)^n \exp(j0.25\pi n)\, u[n]$. The *autocorrelation* sequence $r_{xx}[n]$ corresponding to $x[n]$ is defined as $x[n] * x^*[-n]$. Find (a) $X(z)$, (b) the z-transform of $x^*[-n]$, (c) $R_{xx}(z)$ (by using properties), and (d) $r_{xx}[n]$ from $R_{xx}(z)$.

9. Use z-transform properties to find the convolution result of $x[n]$ and $h[n]$, where $x[n] = \delta[n] - 2\delta[n-2]$ and $h[n] = 2\delta[n] - 2\delta[n-1] + 3\delta[n-2] + \delta[n-3]$.

$X(z) = \sum_{n=-\infty}^{\infty} \{\delta(n) - 2\delta(n-2)\} z^{-n} = 1 - 2z^{-2}$

$H(z) = 2 - 2z^{-1} + 3z^{-2} + z^{-3}$

$Y(z) = X(z)H(z) = \frac{(z^2-2)(2z^3-2z^2+3z+1)}{z^5} = \frac{2z^5-2z^4-z^3+5z^2-6z-2}{z^5}$

$= \{2, -2, -1, 5, -6, -2\}$ (↑ at the first term)

10. Let $x[n] = (0.3)^n \exp(j0.25\pi n)\, u[n]$. The *autocorrelation* sequence $r_{xx}[n]$ corresponding to $x[n]$ is defined as $x[n] * x^*[-n]$. Find (a) $X(z)$, (b) the z-transform of $x^*[-n]$, (c) $R_{xx}(z)$ (by using properties), and (d) $r_{xx}[n]$ from $R_{xx}(z)$.

$x(n) = 0.3^n e^{j\cdot\pi/4 n} u(n) = (0.3e^{j\pi/4})^n u(n)$

a) $X(z) = \frac{1}{1-(0.3e^{j\pi/4})z^{-1}}$ $\quad |z| > 0.3$

b) $x_1(n) = x^*(-n)$ $\quad X_1(z) = \sum_{n=-\infty}^{\infty} x^*(-n) z^{-n}$

$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \Rightarrow X^*(1/z^*) = \sum_{n=-\infty}^{\infty} x^*(n) z^{n} = X_1(z)$

$X_1(z) = \frac{1}{1-0.3e^{j\pi/4}z}$ $\quad |z| < 10/3$

c) $R_{xx}(z) = \frac{1}{1-0.3e^{-j\pi/4}z - 0.3e^{j\pi/4}z^{-1} + 0.09} = \frac{1}{1.09 - 0.3(e^{-j\pi/4}z + e^{j\pi/4}z^{-1})}$

$\frac{3}{10} < |z| < \frac{10}{3}$

EE2004: Digital Signal Processing

Tutorial 5

(March 2, 2026)

1. Using partial fraction expansion find the inverse z-transform of the following:
   (a) Find the $x[n] \in \ell_1$ whose z-transform is
   $$X(z) = \frac{1 - 2z^{-1}}{1 - \frac{5}{2}z^{-1} + z^{-2}}$$
   (b) Find $x[n]$ whose z-transform is
   $$X(z) = \frac{1}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{2}z^{-1})^3} \qquad |z| > \frac{1}{2}$$

2. Using series expansion find the inverse z-transform of
   $$X(z) = \frac{1 + 2z^{-1}}{1 - 2z^{-1} + z^{-2}}$$
   when $x[n]$ is (a) causal, and (b) non-causal.

3. Consider
   $$Y(z) = \frac{1 - a^2}{1 + a^2 - a(z + z^{-1})}$$
   Using long-division, find (i) causal inverse z-transform (quotient should be in powers of z^{-1}), and (ii) anti-causal inverse z-transform (quotient should be in powers of z).

4. The input to a causal LTI system is $x[n] = 2^n u[-n-1] + (1/2)^n u[n]$. The output z-transform is
   $$Y(z) = \frac{-\frac{3}{2}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)(1 - 2z^{-1})}$$
   (a) Determine $H(z)$. Be sure to specify the RoC.
   (b) What is the RoC for $Y(z)$?
   (c) Determine $y[n]$.

5. The following facts are given about and LTI system characterized by $h[n] \overset{\mathcal{Z}}{\longleftrightarrow} H(z)$: (i) $h[n]$ is real-valued, (ii) $h[n]$ is right-sided, (iii) $\lim_{z\to\infty} H(z) = 1$, (iv) $H(z)$ has two zeros, and (v) $H(z)$ has a complex-valued pole on the circle defined by $|z| = 3/4$. Is the system causal? Is it stable?

6. Using the contour integral method, find $x[n]$ for all n when $X(z) = \frac{1}{1 - a\,z^{-1}}$ with RoC $|z| < |a|$.

1. Using partial fraction expansion find the inverse z-transform of the following:

(a) Find the $x[n] \in \ell_1$ whose z-transform is

means sys is stable & has $|z|=1$ in RoC

$$X(z) = \frac{1-2z^{-1}}{1-\frac{5}{2}z^{-1}+z^{-2}}$$

(b) Find $x[n]$ whose z-transform is

$$X(z) = \frac{1}{(1-\frac{1}{4}z^{-1})(1-\frac{1}{2}z^{-1})^3} \qquad |z| > \frac{1}{2}$$

a) $X(z) = \dfrac{z^2-2z}{1-2.5z+z^2} = \dfrac{z(z-2)}{(z-2)(z-0.5)} = \dfrac{1}{1-0.5z^{-1}}$

$\Rightarrow x[n] = 0.5^n u[n]$

b) $X(z) = \dfrac{A}{1-0.25z^{-1}} + \dfrac{B}{1-0.5z^{-1}} + \dfrac{C}{(1-0.5z^{-1})^2} + \dfrac{D}{(1-0.5z^{-1})^3}$

$\Rightarrow A(1-0.5z^{-1})^2 + B(1-0.5z^{-1})^2(1-0.25z^{-1}) = 1$

$+ C(1-0.5z^{-1})(1-0.25z^{-1}) + D(1-0.25z^{-1})$

$z^{-1}=4 \Rightarrow -A = 1 \Rightarrow \boxed{A=-1}$

$z^{-1}=2 \Rightarrow D/2 = 1 \Rightarrow \boxed{D=2}$

$z^{-1}=1 \Rightarrow A/8 + 3B/16 + 3C/8 + 3D/4 = 1 \Rightarrow B/2 + C = -1$

$z^{-1}=-1 \Rightarrow 27A/8 + 45B/16 + 15/8 C + 5D/4 = 1 \Rightarrow 3B/2 + C = 1$

$\boxed{B=2}$ $\boxed{C=-2}$

$x[n] = -(0.5)^n u[n] + 2(0.5)^n u[n] - 2(n+1)(0.5)^n u[n]$

$+ (n+1)(n+2)(0.5)^n u[n]$

2. Using series expansion find the inverse z-transform of

$$X(z) = \frac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$$

$(1-v)^{-2} = \sum_{n=0}^{\infty}(n+1)v^n$

when $x[n]$ is (a) causal, and (b) non-causal.

$X(z) = (1+2z^{-1})(1-z^{-1})^{-2} = \sum_{n=0}^{\infty}(n+1)z^{-n} + \sum_{n=0}^{\infty}2(n+1)z^{-(n+1)}$

a) $(n+1)u[n] + 2n\,u[n-1] = (3n+1)u[n]$

b) $-(3n+1)u[-n-1]$

3. Consider

$$Y(z) = \frac{1-a^2}{1+a^2-a(z+z^{-1})}$$

Using long-division, find (i) causal inverse z-transform (quotient should be in powers of z^{-1}), and (ii) anti-causal inverse z-transform (quotient should be in powers of z).

$Y(z) = \dfrac{z(1-a^2)}{z+za^2-az^2-a} = \dfrac{z(a^2-1)}{az^2-(a^2+1)z+a}$

(i)

$$\begin{array}{l|l} az^2-(a^2+1)z+a & z(a^2-1) \\ & -z(a^2-1)+\frac{(a^4-1)}{a}z^{-1}+(a^2-1)z^{-1} \\ \hline & \frac{a^4-1}{a}-(a^2-1)z^{-1} \end{array}$$

$\dfrac{(a^2-1)}{a}z^{-1} + \dfrac{(a^4-1)}{a^2}z^{-2} + \cdots$

$x[n] = (a^n - a^{-n})u[n]$

(ii)

$$\begin{array}{l|l} az^2-(a^2+1)z+a & z(a^2-1) \\ & -z(a^2-1)+z^2\frac{(a^4-1)}{a}+z^3(a^2-1) \\ \hline & z^2\left(\frac{a^4-1}{a}\right)-z^3(a^2-1) \end{array}$$

$z(a-1/a) + z^2(a^2-1/a^2) + \cdots$

$x[n] = (a^{-n} - a^n)u[-n]$

4. The input to a causal LTI system is $x[n] = 2^n u[-n-1] + (1/2)^n u[n]$. The output z-transform is

$$Y(z) = \frac{-\frac{3}{2}z^{-1}}{\left(1+\frac{1}{2}z^{-1}\right)(1-2z^{-1})}$$

(a) Determine $H(z)$. Be sure to specify the RoC.

(b) What is the RoC for $Y(z)$?

(c) Determine $y[n]$.

$X(z) = \dfrac{-1}{1-2z^{-1}} + \dfrac{1}{1-0.5z^{-1}}$ $\qquad \frac{1}{2} < |z| < 2$

$= \dfrac{-1.5z^{-1}}{(1-2z^{-1})(1-1/2z^{-1})}$

$H(z) = \dfrac{(1-z^{-1}/2)}{(1+z^{-1}/2)}$ $\qquad \frac{1}{2} < |z| < 2$

$Y(z) = \frac{3/5}{1+1/2z^{-1}} - \frac{3/5}{1-2z^{-1}}$ $\quad \frac{3}{5}2^n u[-n-1] + \frac{3}{5}(-1/2)^n u[n]$

5. The following facts are given about and LTI system characterized by $h[n] \overset{\mathcal{Z}}{\longleftrightarrow} H(z)$: (i) $h[n]$ is real-valued, (ii) $h[n]$ is right-sided, (iii) $\lim_{z\to\infty} H(z) = 1$, (iv) $H(z)$ has two zeros, and (v) $H(z)$ has a complex-valued pole on the circle defined by $|z| = 3/4$. Is the system causal? Is it stable?

(i) poles & zeros in conjugate pairs

(ii) ROC: $|z| > |p_{max}|$

(iii) $h(0) = 1$ and causal

(iv) poles: $\frac{3}{4}e^{\pm j\theta}$

since all poles lie inside unit circle, it is stable

6. Using the contour integral method, find $x[n]$ for all n when $X(z) = \frac{1}{1-az^{-1}}$ with RoC $|z| < |a|$.

$X(z) = \frac{z}{z-a}$

$x(n) = \frac{1}{2\pi j}\oint_C X(z) z^{n-1} dz = \frac{1}{2\pi j}\oint_C \frac{z^n}{z-a} dz$

$n>0 \Rightarrow$ one pole at $z=a$ (outside contour C)

$\Rightarrow x(n) = 0 \quad \forall n \geq 0$

$n<0 \Rightarrow n = -m \Rightarrow x(n) = \frac{1}{2\pi j}\oint_C \frac{1}{z^m(z-a)} dz$

$z=0$ pole lies inside contour C

$\Rightarrow x[n] = \text{Res}\left(\frac{1}{z^m(z-a)}, z=0\right)$

for a pole of order m at $z=0$,

$\text{Res}(f,0) = \frac{1}{(m-1)!}\lim_{z\to 0}\frac{d^{m-1}}{dz^{m-1}}\left[\frac{z^m}{z^m(z-a)}\right]$

$= -a^{-m} = -a^n \quad \forall n<0$

$x(n) = -a^n u[-n-1]$

7. Find the sequences whose DTFT are given below: (a) $X(e^{j\omega}) = (\cos\omega)^N$ for an arbitrary integer $N > 0$, and (b) $X(e^{j\omega}) = (\cos\frac{\omega}{2})^N$ for an arbitrary *even* integer $N > 0$.
Hint: Express the cosine function in terms of complex exponentials and apply the binomial theorem.

8. Consider the DTFT $H(e^{j\omega}) = j\omega$ for $\omega \in [-\pi, \pi)$. (a) Determine the corresponding $h[n]$. What is the value of $h[0]$? (b) Apply Parseval's theorem to this transform pair to obtain the value of $\sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1}{n^2}$.

9. Consider the sequence $x[n] = \{2, 3, \underset{\downarrow}{-1}, 0, -4, 3, 1, 2, 4\}$. Find the inverse DTFT of (a) $X(e^{-j\omega})$, and (b) $X(-e^{j\omega})$. Comment on the difference between the two answers.

10. What symmetry property, if any, does the DTFT possess in each of the following cases: (a) $x[n] \in \mathbb{R}$, (b) $x[n] = x[-n]$, in addition to being real-valued, and (c) $x[n] = x^*[-n]$.

11. For each of the following Fourier transforms, use DTFT properties to determine whether or not the corresponding time-domain signal is (i) real, imaginary, or neither, and (ii) even, odd, or neither. Do this without explicitly evaluating the inverse of any of the given transforms.

    (a) $X_1(e^{j\omega}) = e^{-j\omega} \sum_{k=1}^{10} \sin(k\omega)$

    (b) $X_2(e^{j\omega}) = j \sin(\omega)\cos(5\omega)$

    (c) $X_3(e^{j\omega}) = A(\omega) + e^{jB(\omega)}$ where $A(\omega) = \begin{cases} 1, & 0 \le |\omega| \le \frac{\pi}{8} \\ 0, & \frac{\pi}{8} < |\omega| \le \pi \end{cases}$
    and $B(\omega) = -\frac{3\omega}{2} + \pi$

12. Determine $x[n]$ given the following four facts about its DTFT $X(e^{j\omega})$:
    1. $x[n] = 0$ for $n > 0$
    2. $x[0] > 0$
    3. $\mathrm{Im}\{X(e^{j\omega})\} = \sin(\omega) - \sin(2\omega)$
    4. $\frac{1}{2\pi}\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 \, d\omega = 3.$

13. Using DTFT properties determine the numerical value of $A = \sum_{n=0}^{\infty} n \left(\frac{1}{2}\right)^n$. ◇

7. Find the sequences whose DTFT are given below: (a) $X(e^{j\omega}) = (\cos\omega)^N$ for an arbitrary integer $N > 0$, and (b) $X(e^{j\omega}) = (\cos\frac{\omega}{2})^N$ for an arbitrary *even* integer $N > 0$.
Hint: Express the cosine function in terms of complex exponentials and apply the binomial theorem.

a) $X(e^{j\omega}) = \left(\frac{e^{j\omega} + e^{-j\omega}}{2}\right)^N$ $\quad x(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

$= \frac{1}{2^N} \sum_{k=0}^{N} {}^N C_k (e^{j\omega})^k (e^{-j\omega})^{N-k} = \frac{e^{-j\omega N}}{2^N} \sum_{k=0}^{N} {}^N C_k e^{2j\omega k}$

$= \frac{\sum_{k=0}^{N} {}^N C_k e^{-j\omega(N-2k)}}{2^N}$

$\frac{1}{2^N} y(n+N)$ where $y[n] = \{1, 0, {}^N C_1, 0, \ldots {}^N C_N\}$

b) $x[n] = \frac{1}{2^N} y(n + N/2)$ where $y(n) = \{1, {}^N C_1, {}^N C_2, \ldots {}^N C_N\}$

8. Consider the DTFT $H(e^{j\omega}) = j\omega$ for $\omega \in [-\pi, \pi)$. (a) Determine the corresponding $h[n]$. What is the value of $h[0]$? (b) Apply Parseval's theorem to this transform pair to obtain the value of $\sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1}{n^2}$.

$h(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi} (j\omega) e^{j\omega n} d\omega$

$= \frac{1}{2\pi n}\int_{-\pi}^{\pi} \omega \, d(e^{j\omega n}) = \frac{1}{2\pi n} \omega e^{j\omega n}\Big|_{-\pi}^{\pi} - \frac{1}{2\pi n}\int_{-\pi}^{\pi} e^{j\omega n} d\omega$

$= \frac{\pi}{2\pi n}(e^{j\pi n} + e^{j\pi n}) - \frac{1}{2\pi n \cdot jn}(e^{j\pi n} - e^{j\pi n})$

$= \frac{\cos(\pi n)}{n} - \frac{1}{\pi n^2}\sin(\pi n) = \frac{(-1)^n}{n}$ $\quad n \neq 0$

0 $\quad n = 0$

parseval's theorem

$\sum_{n=-\infty}^{\infty} |h[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |H(e^{j\omega})|^2 d\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi} \omega^2 d\omega = \frac{\pi^2}{3}$

$\Rightarrow \sum_{n\neq 0, n=-\infty}^{\infty} \frac{1}{n^2} = \pi^2/3$

9. Consider the sequence $x[n] = \{2, 3, \underset{\downarrow}{-1}, 0, -4, 3, 1, 2, 4\}$. Find the inverse DTFT of (a) $X(e^{-j\omega})$, and (b) $X(-e^{j\omega})$. Comment on the difference between the two answers.

(a) $x_1[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{-j\omega}) e^{j\omega n} d\omega = x(-n)$

(b) $x_2[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j(\omega-\pi)}) e^{j(\omega-\pi)n} e^{j\pi n} d\omega = (-1)^n x[n]$

10. What symmetry property, if any, does the DTFT possess in each of the following cases:
(a) $x[n] \in \mathbb{R}$, (b) $x[n] = x[-n]$, in addition to being real-valued, and (c) $x[n] = x^*[-n]$.

a) $X(e^{j\omega}) = X^*(e^{-j\omega})$ i.e $|X(e^{j\omega})| \rightarrow$ even : $\angle X(e^{j\omega}) \rightarrow$ odd

b) $X(e^{j\omega}) = X(e^{-j\omega})$ $\quad Im(X(e^{j\omega})) = 0$

c) $X(e^{j\omega})$ is purely real

12. Determine $x[n]$ given the following four facts about its DTFT $X(e^{j\omega})$:

1. $x[n] = 0$ for $n > 0$ $\quad$ I $X(e^{j\omega}) =$
2. $x[0] > 0$
3. $\text{Im}\{X(e^{j\omega})\} = \sin(\omega) - \sin(2\omega)$
4. $\frac{1}{2\pi}\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 \, d\omega = 3.$

$j \cdot Im\{X(e^{j\omega})\} \leftrightarrow x_{odd}[n] = \frac{x[n] - x[-n]}{2}$

$\Rightarrow \frac{1}{2}(e^{j\omega} - e^{2j\omega} - e^{-j\omega} + e^{-2j\omega}) \leftrightarrow x_{odd}[n]$

$x_{odd}[-1] = -1/2 \quad x_{odd}[-2] = \frac{1}{2} \Rightarrow x[-1] = -1$
$x[-2] = 1$

$x_{odd}[1] = \frac{1}{2} \quad x_{odd}[2] = -\frac{1}{2}$

$\sum |x[n]|^2 = 3 \Rightarrow x[0] = 1$

13. Using DTFT properties determine the numerical value of $A = \sum_{n=0}^{\infty} n \left(\frac{1}{2}\right)^n$.

$(1/2)^n u[n] \leftrightarrow \frac{1}{1 - \frac{1}{2}e^{j\omega}}$

$n\left(\frac{1}{2}\right)^n u[n] \leftrightarrow j \cdot \frac{d}{d\omega}\left(\frac{1}{1 - \frac{1}{2}e^{j\omega}}\right) = \frac{\frac{1}{2}e^{-j\omega}}{(1 - \frac{1}{2}e^{j\omega})^2}$

$\omega = 0 \Rightarrow \frac{1/2}{(1/2)^2} = 2$

11. For each of the following Fourier transforms, use DTFT properties to determine whether or not the corresponding time-domain signal is (i) real, imaginary, or neither, and (ii) even, odd, or neither. Do this without explicitly evaluating the inverse of any of the given transforms.

(a) $X_1(e^{j\omega}) = e^{-j\omega} \sum_{k=1}^{10} \sin(k\omega)$

(b) $X_2(e^{j\omega}) = j \sin(\omega)\cos(5\omega)$

(c) $X_3(e^{j\omega}) = A(\omega) + e^{jB(\omega)}$ where $A(\omega) = \begin{cases} 1, & 0 \le |\omega| \le \frac{\pi}{8} \\ 0, & \frac{\pi}{8} < |\omega| \le \pi \end{cases}$
and $B(\omega) = -\frac{3\omega}{2} + \pi$

EE2004: Digital Signal Processing

Tutorial 6
(March 9, 2026)

1. Determine and sketch the DTFT $X_i(\omega)$ of $x_i[n]$:
   (a) $x_1[n] = \{1, 1, \underset{\downarrow}{1}, 1, 1\}$
   (b) $x_2[n] = \{1, 0, 1, 0, \underset{\downarrow}{1}, 0, 1, 0, 1\}$
   (c) $x_3[n] = \{1, 0, 0, 1, 0, 0, \underset{\downarrow}{1}, 0, 0, 1, 0, 0, 1\}$
   Is there any relation between $X_1(\omega)$, $X_2(\omega)$, and $X_3(\omega)$?
2. We are given the following information about an LTI system with frequency response $H(e^{j\omega})$: (a) the system is causal, (b) (b) $H(e^{j\omega}) = H^*(e^{-j\omega})$, and (c) the DTFT of $h[n+1]$ is real-valued. Based on the information given, is the filter IIR or FIR? Justify your answer.
3. Consider a discrete-time LTI system with impulse response $h[n] = (1/2)^n u[n]$. Using DTFT properties determine the response to each of the following input signals: (a) $x[n] = (n+1)(1/4)^n u[n]$, and (b) $x[n] = (-1)^n$.
4. Consider a causal LTI system described by the difference equation
$$y[n] + \frac{1}{2}y[n-1] = x[n]$$
Determine the frequency response $H(e^{j\omega})$ of the above system.
5. The input to a cascade of two LTI systems is $x[n] = \cos(0.6\pi n) + 3\delta[n-5] + 2$. The first system is an ideal lowpass filter with $\omega_c = \pi/2$. The second system's input-output relationship is $y[n] = w]n] - w[n-1]$, where $w[n]$ is the output of the first system. Find the output $y[n]$.
6. A causal and stable LTI system accepts the input $\left(\frac{4}{5}\right)^n u[n]$ and produces the output $n\left(\frac{4}{5}\right)^n u[n]$.
   (a) Determine the system's frequency response $H(e^{j\omega})$.
   (b) Determine the system's input-output difference equation.
7. Consider a system obtained by cascading two LTI systems with frequency responses
$$H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + 0.5e^{-j\omega}}$$
and
$$H_2(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega} + 0.25e^{-j2\omega}}$$
(a) Find the difference equation describing the overall system. (b) Determine the impulse response of the overall system.

1. Determine and sketch the DTFT $X_i(\omega)$ of $x_i[n]$:

   (a) $x_1[n] = \{1, 1, \underset{\downarrow}{1}, 1, 1\}$

   (b) $x_2[n] = \{1, 0, 1, 0, \underset{\downarrow}{1}, 0, 1, 0, 1\}$

   (c) $x_3[n] = \{1, 0, 0, 1, 0, 0, \underset{\downarrow}{1}, 0, 0, 1, 0, 0, 1\}$

   Is there any relation between $X_1(\omega)$, $X_2(\omega)$, and $X_3(\omega)$?

a) $X_1(\omega) = 1 + 2(\cos\omega + \cos 2\omega)$

b) $X_2(\omega) = 1 + 2(\cos 2\omega + \cos 4\omega)$

c) $X_3(\omega) = 1 + 2(\cos 3\omega + \cos 6\omega)$

2. We are given the following information about an LTI system with frequency response $H(e^{j\omega})$: (a) the system is causal, (b) (b) $H(e^{j\omega}) = H^*(e^{-j\omega})$, and (c) the DTFT of $h[n+1]$ is real-valued. Based on the information given, is the filter IIR or FIR? Justify your answer.

a) $h(n) = 0 \quad \forall \quad n < 0$

$g(n) = h(n+1)$

$G(e^{j\omega}) \rightarrow$ real $\Rightarrow g(n) = g(-n)$

$\Rightarrow h(n+1) = h(-n+1)$

b) $h(n)$ is real

c) $h[n+1] = h(-n+1)$

$h[n] = \{0, 0, \ldots, 0, \underset{\uparrow}{h[0]}, h[1], h[0], 0, \cdots\} \rightarrow$ FIR

3. Consider a discrete-time LTI system with impulse response $h[n] = (1/2)^n u[n]$. Using DTFT properties determine the response to each of the following input signals: (a) $x[n] = (n+1)(1/4)^n u[n]$, and (b) $x[n] = (-1)^n$.

a) $H(e^{j\omega}) = \dfrac{1}{1 - \frac{1}{2}e^{-j\omega}} \qquad X(e^{j\omega}) = \dfrac{1}{(1 - \frac{1}{4}e^{-j\omega})^2}$

$\Rightarrow Y(e^{j\omega}) = X(e^{j\omega})\, H(e^{j\omega})$

$= \dfrac{4}{1 - \frac{1}{2}e^{-j\omega}} + \dfrac{-1}{(1 - \frac{1}{4}e^{-j\omega})^2} + \dfrac{-2}{1 - \frac{1}{4}e^{-j\omega}}$

$y(n) = 4\left(\frac{1}{2}\right)^n u(n) - 2\left(\frac{1}{4}\right)^n u(n) - (n+1)\left(\frac{1}{4}\right)^n u(n)$

$= \left[4\left(\frac{1}{2}\right)^n - (n+3)\left(\frac{1}{4}\right)^n\right] u(n)$

b) $x(n) = e^{j\omega n} \Rightarrow y(e^{j\omega}) = H(e^{j\omega})\, e^{j\omega n}$

$y(n) = \frac{2}{3}(-1)^n$

4. Consider a causal LTI system described by the difference equation

$$y[n] + \frac{1}{2}y[n-1] = x[n]$$

Determine the frequency response $H(e^{j\omega})$ of the above system.

$Y(e^{j\omega})\left[1 + \frac{1}{2}e^{-j\omega}\right] = X(e^{j\omega}) \Rightarrow H(e^{j\omega}) = \dfrac{1}{1 + \frac{1}{2}e^{-j\omega}}$

5. The input to a cascade of two LTI systems is $x[n] = \cos(0.6\pi n) + 3\delta[n-5] + 2$. The first system is an ideal lowpass filter with $\omega_c = \pi/2$. The second system's input-output relationship is $y[n] = w[n] - w[n-1]$, where $w[n]$ is the output of the first system. Find the output $y[n]$.

$x(n) \xrightarrow{\text{LPF}} 2 + 3\,\dfrac{\sin(0.5\pi(n-5))}{\pi(n-5)} \longrightarrow 3\left(\dfrac{\sin(0.5\pi(n-5))}{\pi(n-5)} - \dfrac{\sin(0.5\pi(n-6))}{\pi(n-6)}\right)$

6. A causal and stable LTI system accepts the input $\left(\frac{4}{5}\right)^n u[n]$ and produces the output $n\left(\frac{4}{5}\right)^n u[n]$.

   (a) Determine the system's frequency response $H(e^{j\omega})$.

   (b) Determine the system's input-output difference equation.

$X(z) = \dfrac{1}{1 - \frac{4}{5}z^{-1}} \qquad Y(z) = \dfrac{\frac{4}{5}z^{-1}}{(1 - \frac{4}{5}z^{-1})^2} \Rightarrow H(z) = \dfrac{\frac{4}{5}z^{-1}}{1 - \frac{4}{5}z^{-1}}$

$Y(z)(1 - 0.8z^{-1}) = X(z)(\frac{4}{5}z^{-1})$

$\Rightarrow y(n) - 0.8y(n-1) = 0.8\,x(n-1)$

7. Consider a system obtained by cascading two LTI systems with frequency responses

$$H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + 0.5e^{-j\omega}}$$

and

$$H_2(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega} + 0.25e^{-j2\omega}}$$

(a) Find the difference equation describing the overall system. (b) Determine the impulse response of the overall system.

a) $H(z) = H_1(z)\,H_2(z) = \dfrac{2 - z^{-1}}{1 + 0.125z^{-3}} \Rightarrow y(n) + \frac{1}{8}y(n-3) = 2x(n) - x(n-1)$

b) $\dfrac{1}{1 + 0.125z^{-3}} = \sum_{k=0}^{\infty}(-0.125z^{-3})^k = \sum_{k=0}^{\infty}(-0.125)^k z^{-3k}$

$\Rightarrow z^{-1}\left\{\dfrac{1}{1 + 0.125z^{-3}}\right\} = (-0.125)^k \quad$ for $n = 3, 6, 9, \ldots$

$z^{-1}\left\{\dfrac{z^{-1}}{1 + 0.125z^{-3}}\right\} = (-0.125)^k \quad$ for $n = 1, 4, 7, \ldots$

$h(n) = 2(-0.5)^n \delta[n \bmod 3] - (0.5)^{n-1} \delta[(n-1) \bmod 3]$

8. Consider and LTI system with frequency response $H(e^{j\omega}) = \dfrac{1 - e^{-j2\omega}}{1 + r^2 e^{-j2\omega}}$ where $r = 0.9$. The input to this system is $\cos n\pi/8 + \cos n\pi/2$. The output can be expressed as $A_1 \cos(n\pi/8 + \theta_1) + A_2 \cos(n\pi/2 + \theta_2)$. Determine A_i and θ_i for $i = 1, 2$.

$H(e^{j \cdot \pi/2}) = \frac{2}{1 - r^2} = 10.526 = A_2 \qquad \theta_2 = 0$

$H(e^{j\pi/8}) = 0.457 \angle 87.5^\circ$ ($\to A_1$, $\to \theta_1$)

8. Consider and LTI system with frequency response $H(e^{j\omega}) = \dfrac{1 - e^{-j2\omega}}{1 + r^2 e^{-j2\omega}}$ where $r = 0.9$. The input to this system is $\cos n\pi/8 + \cos n\pi/2$. The output can be expressed as $A_1 \cos(n\pi/8 + \theta_1) + A_2 \cos(n\pi/2 + \theta_2)$. Determine A_i and θ_i for $i = 1, 2$.

9. **Computer assignment** Let $x[n] = \{1, \underset{\downarrow}{2}, 3, 4, 5\}$. Numerically evaluate the sequence's DTFT at 501 equispaced points over the interval $[0, \pi]$.

10. **Computer assignment** Numerically verify the validity of the shift property as follows: (i) create a vector of length 11 with random entries using the `rand` function; (ii) compute its DTFT at 501 equispaced points over the interval $[0, \pi]$; (iii) evaluate numerically $e^{-j2\omega} X(e^{j\omega})$ over the same set of points; (iv) form a shifted sequence by adding two zeros at the beginning of the vector created in (i); (v) compute the shifted vector's DTFT at 501 equispaced points over the interval $[0, \pi]$; (vi) finally compute the maximum absolute error of the difference vector and confirm that the value is extremely small.

11. **Computer assignment** Consider the LTI system specified by the difference equation $y[n] = 0.8\, y[n-1] + x[n]$. Let the input to this system be $x[n] = \cos(0.05\pi n)\, u[n]$.
    (a) Generate 100 samples of the input $x[n]$ and store it in the vector `x`.
    (b) Understand MATLAB's `filter` command and obtain the output `y` using `y = filter(b,a,x);` (the vectors `b` and `a` contain the numerator and denominator coefficients of the rational transfer function $H(e^{j\omega})$).
    (c) Plot `x` and `y` in a figure one below the other (use the command `subplot`). Observe the initial cycles and compare them with the later ones; what are the main differences? Can you explain why the initial part is different?

◇

EE2004: Digital Signal Processing

Tutorial 7

(March 17, 2026)

1. Consider the filter $H(z)$ with difference equation $y[n] - y[n-2] = x[n] - x[n-6]$.
   (a) Determine the poles and zeros of $H(z)$. Is $H(z)$ a causal and stable filter?
   (b) Derive the magnitude response of $H(z)$, and sketch it for $0 \le \omega \le \pi$.

2. Let
$$H(z) = \frac{1}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$
   (a) Show that the peak of $|H(e^{j\omega})|$ occurs at $\omega_0 = \cos^{-1}\left(\frac{1+r^2}{2r}\cos\theta\right)$.
   (b) Calculate the theoretical peak location when $r = 0.8$ and $\theta = \pi/10$ and verify it using MATLAB.

3. Consider the causal and stable LTI filter with transfer function $H(z) = \frac{1}{1 - a\,z^{-1}}$, where $0 < a < 1$. Find the closed-form expression for the frequency at which the magnitude-squared transfer function is half the peak value.

4. Consider the improved bandpass filter with zeros at $z = \pm 1$:
$$H_{\text{BP}}(z) = \frac{1-\alpha}{2} \cdot \frac{1 - z^{-2}}{1 - \beta(1+\alpha)z^{-1} + \alpha z^{-2}}$$
   The poles are at $re^{\pm j\theta}$, and hence $r = \sqrt{\alpha}$ and $2\sqrt{\alpha}\cos\theta = \beta(1+\alpha)$.
   (a) Show that the magnitude-squared response can be written as
$$\frac{(1-r^2)^2 \sin^2\omega}{(1+r^2)^2(\cos\omega - \beta)^2 + (1-r^2)^2 \sin^2\omega}$$
   (b) Observe $|H_{\text{BP}}(e^{j\omega})|^2 = K\, y^2/(x^2+y^2)$ where x, y are functions of ω and K is a constant. Show that the *maximum* occurs when $x = 0$. Hence determine the value of ω in terms of r and θ at which this maximum is attained.

5. **Computer assignment** Plot the unwrapped phase angle for a stable all-pass filter. Verify its magnitude response is unity for all frequencies. Try various orders and pole locations (once you choose the poles, the zeros are automatically fixed). Does the phase function exhibit any particular characteristic? What are the values of the phase at 0, π, and 2π? Is the phase angle at $\omega = 2\pi$ related to the filter order?

6. **Computer assignment** Compute and plot the phase response of $H(z) = [(1+z^{-1})/2]^6$.
   (a) Determine analytically $\angle H(e^{j\omega})$ and using this expression plot the phase response.
   (b) Use MATLAB's `freqz` command and obtain the phase response and compare it with the results obtained from the previous plot.

1. Consider the filter $H(z)$ with difference equation $y[n] - y[n-2] = x[n] - x[n-6]$.
   (a) Determine the poles and zeros of $H(z)$. Is $H(z)$ a causal and stable filter?
   (b) Derive the magnitude response of $H(z)$, and sketch it for $0 \le \omega \le \pi$.

a) $Y(z) - Y(z)\cdot z^{-2} = X(z) - X(z) z^{-6}$

$\frac{Y(z)}{X(z)} = \frac{z^6 - 1}{z^2 - 1}\cdot\frac{z^2}{z^6} = \frac{1}{z^4}(z^4 + z^2 + 1)$

poles: 0

zeros: $\pm\frac{1}{2} \pm \frac{\sqrt{3}j}{2}$,

b) $H(e^{j\omega}) = e^{4j\omega} + e^{2j\omega} + 1 = (\cos 4\omega + \cos 2\omega + 1) - j(\sin 4\omega + \sin 2\omega)$

$|H(e^{j\omega})| = \sqrt{1+1+1+2\cos 4\omega + 2\cos 2\omega + 2\cos 2\omega}$

$= \sqrt{3 + 2\cos 4\omega + 4\cos 2\omega} = |2\cos 2\omega + 1|$

2. Let
$$H(z) = \frac{1}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$
   (a) Show that the peak of $|H(e^{j\omega})|$ occurs at $\omega_0 = \cos^{-1}\left(\frac{1+r^2}{2r}\cos\theta\right)$.
   (b) Calculate the theoretical peak location when $r = 0.8$ and $\theta = \pi/10$ and verify it using MATLAB.

(a) $H(e^{j\omega}) = \frac{1}{1 - 2r\cos\theta\, e^{-j\omega} + r^2 e^{-2j\omega}}$

$= \frac{1}{1 - 2r\cos\theta(\cos\omega - j\sin\omega) + r^2(\cos 2\omega - j\sin 2\omega)}$

$= \frac{1}{(1 - 2r\cos\theta\cos\omega + r^2\cos 2\omega) + j(2r\cos\theta\sin\omega - r^2\sin 2\omega)}$

$|H(e^{j\omega})| = \frac{1}{\sqrt{(1 - 2r\cos\theta\cos\omega + r^2\cos 2\omega)^2 + (2r\cos\theta\sin\omega - r^2\sin 2\omega)^2}}$

$1 + 4r^2\cos^2\theta + r^4 - 4r\cos\theta\cos\omega + 2r^2\cos 2\omega - 4r^3\cos\theta\cos\omega$

$\omega_{\min}$

$\Rightarrow (4r^2)\cos^2\omega - (4r\cos\theta + 4r^3\cos\theta)\cos\omega + (r^4 - 2r^2\cos 2\theta + 1)$

$\cos\omega = \frac{4r\cos\theta + 4r^3\cos\theta}{8r^2} = \left(\frac{1+r^2}{2r}\cos\theta\right)$

b) $\cos\omega = \left(\frac{1.64}{1.6}\cos\frac{\pi}{10}\right) = 0.974832$

$\Rightarrow \omega = 0.0716\pi$

3. Consider the causal and stable LTI filter with transfer function $H(z) = \frac{1}{1-az^{-1}}$, where $0 < a < 1$. Find the closed-form expression for the frequency at which the magnitude-squared transfer function is half the peak value.

$H(e^{j\omega}) = \frac{1}{1-ae^{-j\omega}}$

$|H(e^{j\omega})|^2 = \frac{1}{1+a^2-2a\cos\omega}$

$\text{peak}^2 = \frac{1}{(1-a)^2}$

$\frac{1}{1+a^2-2a\cos\omega} = \frac{1}{2(1-a)^2} \Rightarrow \cos\omega = \left(2 - \frac{1}{2}\left(a + \frac{1}{a}\right)\right)$

4. Consider the improved bandpass filter with zeros at $z = \pm 1$:

$$H_{\text{BP}}(z) = \frac{1-\alpha}{2} \cdot \frac{1-z^{-2}}{1-\beta(1+\alpha)z^{-1}+\alpha z^{-2}}$$

The poles are at $re^{\pm j\theta}$, and hence $r = \sqrt{\alpha}$ and $2\sqrt{\alpha}\cos\theta = \beta(1+\alpha)$. $\cos\theta = \frac{\beta(1+r^2)}{2r}$

(a) Show that the magnitude-squared response can be written as

$$\frac{(1-r^2)^2\sin^2\omega}{(1+r^2)^2(\cos\omega-\beta)^2+(1-r^2)^2\sin^2\omega}$$

(b) Observe $|H_{\text{BP}}(e^{j\omega})|^2 = K\,y^2/(x^2+y^2)$ where x, y are functions of ω and K is a constant. Show that the *maximum* occurs when $x = 0$. Hence determine the value of ω in terms of r and θ at which this maximum is attained.

a) $|H(e^{j\omega})| = \frac{1-r^2}{2}\frac{e^{2j\omega}-1}{e^{2j\omega} - 2r\cos\theta e^{j\omega} + r^2}$

$= \frac{1-r^2}{2}\frac{(\cos 2\omega - 1) + j(\sin 2\omega)}{(\cos 2\omega - 2r\cos\theta\cos\omega + r^2) + j(\sin 2\omega - 2r\cos\theta\sin\omega)}$

$|H(e^{j\omega})|^2 = \left(\frac{1-r^2}{2}\right)^2 \frac{2-2\cos 2\omega}{r^4+1+4r^2\cos^2\theta - 4r^3\cos\theta\cos\omega + 2r^2\cos 2\omega - 4r\cos\theta\cos\omega}$

$= \frac{(1-r^2)^2\sin^2\omega}{r^4+1+4r^2\cos^2\theta - 4r^3\cos\theta\cos\omega + 2r^2\cos 2\omega - 4r\cos\theta\cos\omega}$

$= \frac{(1-r^2)^2\sin^2\omega}{r^4+1+\frac{4r^2\beta(1+r^2)^2}{4r^2} - 4r^3\cos\omega\,\beta\left(\frac{1+r^2}{2r}\right) + 2r^2\cos 2\omega - 2\cancel{2r}\cdot\frac{\beta(1+r^2)}{\cancel{2r}}\cos\omega}$

$(1+r^2)^2(\cos\omega-\beta)^2 + (1-r^2)^2\sin^2\omega$

$(1+r^2)^2\beta^2 - 2\beta\cos\omega(1+r^2)^2 + \cos^2\omega(1+r^2)^2 + (1-r^2)^2\sin^2\omega$

$= \frac{(1-r^2)^2\sin^2\omega}{(1+r^2)^2(\cos\omega-\beta)^2 + (1-r^2)^2\sin^2\omega}$

b) $y = (1-r^2)\sin\omega \qquad |H(e^{j\omega})|^2 = \frac{y^2}{x^2+y^2} \le 1$

$x = (1+r^2)(\cos\omega - \beta)$

if $x = 0 \Rightarrow \cos\omega = \beta$

$\cos\omega = \frac{2r\cos\theta}{1+r^2} \Rightarrow \omega = \cos^{-1}\left(\frac{2r\cos\theta}{1+r^2}\right)$

EE2004: Digital Signal Processing

Tutorial 8
(March 26, 2026)

1. Consider a first order, causal, and stable allpass filter of the form

$$A_1(z) = \frac{1 - c_1^* z}{z - c_1}$$

Determine the expression for $1 - |A_1(z)|^2$ and then show that

$$1 - |A_1(z)|^2 \begin{cases} < 0 & \text{for } |z|^2 < 1 \\ = 0 & \text{for } |z|^2 = 1 \\ > 0 & \text{for } |z|^2 > 1 \end{cases}$$

From this deduce that the above property holds for an N-th order allpass filter.

2. In the previous problem, how do the inequalities in the expression for $1 - |A_1(z)|^2$ change if

$$A_1(z) = \frac{-c_1^* + z^{-1}}{1 - c_1 z^{-1}}$$

3. Consider the causal sequence $x[n]$ with z-transform

$$X(z) = \frac{\left(1 - \frac{1}{2} z^{-1}\right)\left(1 - \frac{1}{4} z^{-1}\right)\left(1 - \frac{1}{5} z\right)}{\left(1 - \frac{1}{6} z\right)}$$

For what values of a is $a^n\, x[n]$ a real-valued, minimum-phase sequence, i.e., has all its poles and zeros inside the unit circle ?

4. Eight finite-duration sequences are given below, all of which have identical magnitude frequency response (to within a scale factor). Which of the sequences has all of its zeros *inside* the unit circle?

(a) $\{20, -61, 46, -8\}$
(b) $\{32, -112, 106, -20\}$
(c) $\{-4, 29, -62, 40\}$
(d) $\{15, -102, 192, -96\}$
(e) $\{8, -46, 61, -20\}$
(f) $\{-32, 64, -34, 5\}$
(g) $\{10, -53, 56, -16\}$
(h) $\{200, -310, 145, -20\}$

5. Compute the "partial energy" of the sequences using the formula $E_k = \sum_{n=0}^{k} |h[n]|^2$ for $k = 0, \ldots, 3$ and normalize it by the sequences "total energy", viz., E_3. Can you identifiy the minimum phase sequences from partial energy sequences? From this it should be clear that "minimum energy delay sequence" is a more apt term than "minimum phase sequence". Where are the zeros of the sequence that has the maximum energy delay ?

7. **Computer assignment** Consider the notch filter

$$H(z) = \frac{1 - 2\cos\theta\, z^{-1} + z^{-2}}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$

Let $\theta = \pi/3$. Plot the magnitude frequency response for $r = 0.9$, $r = 0.95$, $r = 0.99$, and $r = 0.999$. ◇

1. Consider a first order, causal, and stable allpass filter of the form

$$A_1(z) = \frac{1 - c_1^* z}{z - c_1}$$ → pole @ c_1 and stable ⇒ $|c_1|<1$

Determine the expression for $1 - |A_1(z)|^2$ and then show that

$$1 - |A_1(z)|^2 \begin{cases} < 0 & \text{for } |z|^2 < 1 \\ = 0 & \text{for } |z|^2 = 1 \\ > 0 & \text{for } |z|^2 > 1 \end{cases}$$

$$|A_1(z)|^2 = \frac{1 - c_1^* z}{z - c_1} \cdot \frac{1 - c_1 z^*}{z^* - c_1^*} = \frac{1 - c_1^* z - c_1 z^* + |z c_1|^2}{|z|^2 + |c_1|^2 - z c_1^* - z^* c_1}$$

$$\frac{1 - c_1^* z - c_1 z^* + |z c_1|^2 + |z|^2 + |c_1|^2 - |z|^2 - |c_1|^2}{|z|^2 + |c_1|^2 - z c_1^* - z^* c_1}$$

$$= 1 + \frac{|z c_1|^2 - |z|^2 - |c_1|^2 + 1}{|z|^2 + |c_1|^2 - z c_1^* - z^* c_1}$$

$$\Rightarrow 1 - |A_1(z)|^2 = \frac{-(|z|^2 - 1)(|c_1|^2 - 1)}{|z - c_1|^2}$$ (numerator < 0, denominator > 0)

2. In the previous problem, how do the inequalities in the expression for $1 - |A_1(z)|^2$ change if

$$A_1(z) = \frac{-c_1^* + z^{-1}}{1 - c_1 z^{-1}}$$

$$|A_1(z)|^2 = A_1(z) A_1^*(z) = \frac{-c_1^* + z^{-1}}{1 - c_1 z^{-1}} \cdot \frac{-c_1 + (z^{-1})^*}{1 - c_1^* (z^{-1})^*}$$

$$= \frac{|c_1|^2 + |z^{-1}|^2 - c_1 z^{-1} - c_1^* (z^{-1})^*}{1 - c_1 z^{-1} - c_1^* (z^{-1})^* + |c_1 z^{-1}|^2}$$

$$= \frac{1 - c_1 z^{-1} - c_1^* (z^{-1})^* + |c_1 z^{-1}|^2 - 1 - |c_1 z^{-1}|^2 + |z^{-1}|^2 + |c_1|^2}{1 - c_1 z^{-1} - c_1^* (z^{-1})^* + |c_1 z^{-1}|^2}$$

$$= 1 - \frac{(|c_1|^2 - 1)(|z^{-1}|^2 - 1)}{(1 - c_1 z^{-1})^2}$$

$$\Rightarrow 1 - |A_1(z)|^2 = \frac{(|c_1|^2 - 1)(|z^{-1}|^2 - 1)}{|1 - c_1 z^{-1}|^2}$$ (−ve / +ve)

3. Consider the causal sequence $x[n]$ with z-transform

$$X(z) = \frac{\left(1 - \frac{1}{2} z^{-1}\right)\left(1 - \frac{1}{4} z^{-1}\right)\left(1 - \frac{1}{5} z\right)}{\left(1 - \frac{1}{6} z\right)}$$

For what values of a is $a^n x[n]$ a real-valued, minimum-phase sequence, i.e., has all its poles and zeros inside the unit circle?

→ $x(z/a)$

$$X(z/a) = \frac{\left(1 - \frac{1}{2}\frac{a}{z}\right)\left(1 - \frac{1}{4}\frac{a}{z}\right)\left(1 - \frac{1}{5}\frac{z}{a}\right)}{\left(1 - \frac{1}{6}\cdot\frac{z}{a}\right)}$$

zeros: $a/2, a/4, 5a$

poles: $6a$

$|a|<2$; $|a|<4$; $|a|<1/5$; $|a|<\frac{1}{6}$ ⇒ $\boxed{|a|<1/6}$; $a \neq 0$

4. Eight finite-duration sequences are given below, all of which have identical magnitude frequency response (to within a scale factor). Which of the sequences has all of its zeros *inside* the unit circle?

(a) {20, −61, 46, −8}
(b) {32, −112, 106, −20}
(c) {−4, 29, −62, 40}
(d) {15, −102, 192, −96}
(e) {8, −46, 61, −20}
(f) {−32, 64, −34, 5}
(g) {10, −53, 56, −16}
(h) {200, −310, 145, −20}

min phase → energy conc early
max phase → energy conc late
same m.R → time reversed

Larger magnitudes in the beginning and decay towards the end

5. Compute the "partial energy" of the sequences using the formula $E_k = \sum_{n=0}^{k} |h[n]|^2$ for $k = 0, \ldots, 3$ and normalize it by the sequences "total energy", viz., E_3. Can you identify the minimum phase sequences from partial energy sequences? From this it should be clear that "minimum energy delay sequence" is a more apt term than "minimum phase sequence". Where are the zeros of the sequence that has the maximum energy delay?

min-phase seq → energy comes early ⇒ E_k/E_3 grows fast
↳ min energy delay

max-phase seq → energy comes late ⇒ E_k/E_3 grows slow (most energy @ end) → max energy delay

min energy delay sequence : h (|zeros| < 1)

max energy delay sequence : c z = 1.25, 2, 4

6. **Computer assignment** Consider the sequences given in Question 4. Plot the magnitude frequency responses of each sequence on a log-magnitude scale. If they have the same shape but are shifted versions of one another, find out the scale factors needed to make all the magnitude responses identical.

7. **Computer assignment** An FIR filter using the Parks-McClellan algorithm can be designed using the command `h = firpm(32,[0 0.3 0.4 1],[1 1 0 0])`. Let `H = freqz(h, 1, 4000)`. Plot magnitude in both linear and log scales and observe the differences. Plot the unwrapped phase angle and observe its properties. Plot `zplane (roots(h))` and verify that jumps of π occur in the phase response at unit circle zeros locations. Let $x[n] = \cos(\omega_1 n)$, where $\omega_1/2\pi = 0.1$. Filter this sequence through the above filter by convolving the input with the impulse response. By suitably delaying the input are you able to align it exactly with the output? Ignore the transients at the beginning and at the end. Is this delay related to the length of the filter? Is it related to the slope of the phase response? Now reduce the length of the filter by one by changing the first argument of `firpm`. Are you now able to align exactly the input and the filtered output? What is the slope of the phase response? Explain your answer.

8. **Computer assignment** Let us compare how an IIR filter with almost identical magnitude response, but whose phase is highly nonlinear, behaves when an input with two tones is applied to it. An *elliptic filter* can be designed using the command `[b,a] = ellip(5,0.3,35,0.3)`. Compare the magnitude response of this filter with the one designed in the previous example. Let us consider the response to $x[n] = \cos(\omega_1 n) + \cos(\omega_2 n)$, where ω_1 is as before and $\omega_2/2\pi = 0.15$. Note that the second component is at the band edge, where the magnitude is still close to unity for both filters, but the phase response changes rapidly for the elliptic filter. Plot the group delay using `grpdelay(b, a, 4000)`. How different are the values for the two tones? Compute the output using `filter(b, a, x)` and plot the result. Is it possible to align the elliptic filter's input and output by suitably delaying one of them? Carefully contrast this filter's behaviour (nonlinear phase response) with that of its counterpart in the previous problem (exactly linear phase response). Add a third tone with frequency $\omega_3/2\pi = 0.4$ to the input and verify that both filters attenuate it significantly, illustrating frequency selective filtering. You can also add noise (use the command `randn`) and see how the components that fall in the stopband get attenuated.

9. **Computer assignment** Compute and plot the phase response using the functions `freqz`, `angle`, `phasez`, `unwrap`, and `phasedelay` for the following systems:
   (a) The pure delay system given by $y[n] = x[n-15]$
   (b) $H(z) = \dfrac{1 + 1.655\,z^{-1} + 1.655\,z^{-2} + z^{-3}}{1 - 1.57\,z^{-1} + 1.264\,z^{-2} - 0.4\,z^{-3}}$.

10. **Computer assignment** Compute the group delay of the system

$$H(z) = \frac{1 + z^{-1} + z^{-2} + z^{-3}}{1 + 0.9\,z^{-1} + 0.81\,z^{-2} + 0.927\,z^{-3}}$$

using the MATLAB function `grpdelay`.

EE2004: Digital Signal Processing

Tutorial 9
(April 9, 2026)

1. Let $H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n]\,e^{-j\omega n}$ where $h[n] = h_R[n] + j\,h_I[n]$, i.e., the filter coefficients $h[n]$ are complex-valued. It is given that the filter is linear phase. Derive the condition on the impulse response imposed by this constraint.

2. Both $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ are generalized linear phase systems. Which of the following systems must also belong to the same class? (a) $H_1(e^{j\omega}) + H_2(e^{j\omega})$, and (b) $H_1(e^{j\omega}) \cdot H_2(e^{j\omega})$.

3. A Type I FIR filter with real-valued coefficients has impulse response length $N = 9$. Its known zeros are $z_1 = -0.5$, $z_2 = 0.3 + j0.5$, and $z_3 = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$. Determine the location of the remaining zeros and find the expression for $H(z)$.

4. Let $H(z)$ be a Type I linear phase lowpass filter of length $N(= M-1)$; $H_{ZP}(\omega) \approx 1$ for $0 \le \omega \le \pi/4$, and $H_{ZP}(\omega) \approx 0$ for $\pi/3 \le \omega \le \pi$. Another filter $G(z)$ is obtained from $H(z)$ as $G(z) = z^{-M/2} - H(z)$. [Note: The notation $H_{ZP}(\omega)$ denotes the zero-phase response.]
   (a) Express the impulse response $g[n]$ in terms of $h[n]$. To what linear phase filter type does $g[n]$ belong?
   (b) Obtain an expression for $G_{ZP}(\omega)$ in terms of $H_{ZP}(\omega)$, and give a rough sketch of $G_{ZP}(\omega)$ for $0 \le \omega \le \pi$.
   (c) Compare the behaviour of $G_{ZP}(\omega)$ with that of $F_{ZP}(\omega)$ where $F(z)$ equals $(-1)^{M/2}H(-z)$.

5. Derive the closed-form expressions for $H_{ZP}(\omega)$ if $H(z)$ equals (a) $(1 + z^{-1})^2(1 - z^{-1})$, and (b) $(1 + z^{-2})(1 - z^{-1})^2$. For both these cases, derive and sketch $\angle H(e^{j\omega})$ over the interval $0 \le \omega \le \pi$.

6. Design a Type III linear phase filter with $M = 4$ satisfying the following conditions: $H_{ZP}(0.3\pi) = 0.2$ and $H_{ZP}(0.6\pi) = 0.8$.

7. Let $h[n]$ be the impulse response of a causal, linear phase FIR filter. We are given the following facts:
   (a) $\angle H(e^{j0+}) = -3\omega + \pi/2$.
   (b) $H(e^{j\pi}) = 0$.
   (c) $\dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} |H(e^{j\omega})|^2\,d\omega = 28$.
   (d) Let $y[n] = h[n] * u[n]$. $y[1] = 4$ and $y[2] = 6$.
   (e) $\dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} \mathrm{Re}\{H(e^{j\omega})\}\,e^{j6\omega}\,d\omega = -1.5$.

   Find $h[n]$.

1. Let $H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n]\, e^{-j\omega n}$ where $h[n] = h_R[n] + j\, h_I[n]$, i.e., the filter coefficients $h[n]$ are complex-valued. It is given that the filter is linear phase. Derive the condition on the impulse response imposed by this constraint.

$H(e^{j\omega}) = A(\omega)\, e^{j(\beta - \alpha\omega)}$

$H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=0}^{N-1} h[n]\, e^{j\omega\left(\frac{N-1}{2} - n\right)}$

$H^*(e^{j\omega}) = \sum_{n=0}^{N-1} h^*[n] e^{j\omega n}$

$H(e^{j\omega}) = e^{-j\omega(N-1)}\, H^*(e^{j\omega}) = \sum_{n=0}^{N-1} h^*[n] e^{-j\omega(N-1-n)}$

$h[n] = h^*[N-1-n]$

real → even symmetric

img → odd symmetric

2. Both $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ are generalized linear phase systems. Which of the following systems must also belong to the same class? (a) $H_1(e^{j\omega}) + H_2(e^{j\omega})$, and (b) $H_1(e^{j\omega}) \cdot H_2(e^{j\omega})$.

$H_1 + H_2 \rightarrow$ no $H_1 \cdot H_2 \rightarrow$ yes

3. A Type I FIR filter with real-valued coefficients has impulse response length $N = 9$. Its known zeros are $z_1 = -0.5$, $z_2 = 0.3 + j0.5$, and $z_3 = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$. Determine the location of the remaining zeros and find the expression for $H(z)$.

2, 1/2

$H(z) = h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3} + h(4)z^{-4} + h(3)z^{-5} + h(2)z^{-6} + h(1)z^{-7} + h(0)z^{-8}$

$H(z) = \dfrac{h(0)z^8 + h(1)z^7 + h(2)z^6 + h(3)z^5 + h(1)z^4 + h(3)z^3 + h(2)z^2 + h(1)z + h(0)}{z^8}$

zeros : $-0.5, -2, 0.3+j0.5, \frac{30-50j}{34}, -\frac{1}{2}+j\frac{\sqrt{3}}{2}, -\frac{1}{2}-j\frac{\sqrt{3}}{2}, 0.3-j0.5, \frac{30+50j}{34}$

(reciprocals & conjugate)

4. Let $H(z)$ be a Type I linear phase lowpass filter of length $N(= M-1)$; $H_{zp}(\omega) \approx 1$ for $0 \le \omega \le \pi/4$, and $H_{zp}(\omega) \approx 0$ for $\pi/3 \le \omega \le \pi$. Another filter $G(z)$ is obtained from $H(z)$ as $G(z) = z^{-M/2} - H(z)$. [Note: The notation $H_{zp}(\omega)$ denotes the zero-phase response.]

   (a) Express the impulse response $g[n]$ in terms of $h[n]$. To what linear phase filter type does $g[n]$ belong?

   (b) Obtain an expression for $G_{zp}(\omega)$ in terms of $H_{zp}(\omega)$, and give a rough sketch of $G_{zp}(\omega)$ for $0 \le \omega \le \pi$.

   (c) Compare the behaviour of $G_{zp}(\omega)$ with that of $F_{zp}(\omega)$ where $F(z)$ equals $(-1)^{M/2} H(-z)$.

(a) $g(n) = \delta(n - M/2) - h(n)$ → type I as it's symmetric about $n = M/2$, M being even

(b) for a linear phase system with C.O.S @ M/2

$H(e^{j\omega}) = H_{zp}(\omega)\, e^{-j\omega M/2}$

$G(e^{j\omega}) = e^{-j\omega M/2} - H(e^{j\omega}) = e^{-j\omega M/2} - H_{zp}(\omega) e^{-j\omega M/2}$

$\Rightarrow G_{zp}(\omega) = 1 - H_{zp}(\omega) \Rightarrow$ HPF

(c) $F(e^{j\omega}) = (-1)^{M/2} H(-e^{j\omega}) = (-1)^{M/2} H_{zp}(\omega - \pi)\, e^{-j(\omega-\pi)M/2}$

$= H_{zp}(\omega - \pi)\, e^{-j\omega M/2}$

$\Rightarrow F_{zp}(\omega) = H_{zp}(\omega - \pi) = H_{zp}(\pi - \omega)$

EE2004: Digital Signal Processing

Tutorial 10
(April 16, 2026)

1. Find the Nyquist sampling rates for the following signals: (a) $\text{sinc}(20t)$, (b) $4\,\text{sinc}^2(100t)$, (c) $8\,\sin(50\pi t)$, (d) $4\,\sin(30\pi t) + 3\,\cos(70\pi t)$, (e) $\text{rect}(300t)$.

2. Let $x_c(t) = \cos(4000\pi t)$.
   (a) If the signal is sampled every $T = 1/6000$ s, will there be any aliasing? Sketch $X_s(F)$.
   (b) If the signal is sampled every $T = 1/1500$ s, will there be any aliasing? Sketch $X_s(F)$.

3. The signal $x_c(t) = \cos(2\pi F_0 t + \phi)$ is sampled at F_s Hz where $F_0 = 8$ kHz and $F_s = 12$ kHz, giving rise to $x[n] = \cos(\omega_0 n + \theta)$. It is given that $0 < \omega_0 < \pi$. Determine the value of (i) ω_0 and (ii) θ in terms of ϕ.

4. The sequence $\phi[n] = \cos\left(\frac{\pi}{4}n\right)$, was obtained by sampling a continuous-time signal $\phi(t) = \cos(\omega_0 t)$, at a sampling rate of 1000 samples/s. What are two possible positive values of ω_0 that could have resulted in the sequence $\phi[n]$?

5. The continuous-time signal $\phi(t) = 4\sin(20\pi t) - 5\cos(24\pi t) + 3\sin(120\pi t) + 2\cos(176\pi t)$ is sampled at $50Hz$ rate, generating the sequence $\phi[n]$. Determine the exact expression of $\phi[n]$.

6. Let $x(t) = \sum_{-10}^{10} jk\, e^{j16\pi kt}$. Find Nyquist sampling rate.

7. The spectrum $X(F) \in \mathbb{R}$ of a certain continuous-time signal is triangular in shape centred at $F = 0$ with $X(0) = 1$ and $X(F) = 0$ for $|F| > F_c$. The underlying $x(t)$ is sampled at F_s Hz, with $x_s(t)$ being its associated impulse-train sampled signal and $x[n]$ being the corresponding discrete-time sequence. Find $X_s(F)$, $X_s(j\Omega)$, $X(e^{j\omega})$, and $X(e^{j2\pi F})$ when (a) $2F_c < F_s$, and (b) $2F_c > F_s$.

8. Let $x(t)$ be a continuous time signal having Nyquist frequency Ω_c. Find Nyquist frequencies of the following signals in terms of Ω_c: (a) $x_1(t) = x^2(t)$, (b) $x_2(t) = x(t/5)$, (c) $x_3(t) = x(4t)$, (d) $x_4(t) = \int_{-\infty}^{\infty} x(t-\tau)x(\tau)\,d\tau$, and (e) $x_5(t) = \dfrac{dx(t)}{dt}$.

9. The continuous-time signal $x_c(t) = \sin(20\pi t) + \cos(40\pi t)$ is sampled with a sampling period T to obtain the discrete-time signal $x[n] = \sin\left(\dfrac{\pi n}{5}\right) + \cos\left(\dfrac{2\pi n}{5}\right)$.
   (a) Determine a choice for T consistent with this information.
   (b) If your choice of T is unique, explain why. If not, specify another choice of T consistent with this information.

10. **Computer assignment** The ideal reconstruction filter implements the following computation:

$$x_c(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$

which is non-realizable because it requires an infinite number of samples. Instead consider,

$$\hat{x}_c(t) = \sum_{n=0}^{N} x[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$

which uses only $(N+1)$ terms. We will study the resulting error $e(t) = x_c(t) - \hat{x}_c(t)$ in the interval $0 \le t \le NT$. Most observations can be made with $3 \le N \le 20$.
   (a) Develop a MATLAB function `[xhat,t] = DAC(x,N,T)` that implements the finite sum over the interval $0 \le t \le NT$ given the samples $x[n]$ over $0 \le n \le N$ and the sampling interval T. To obtain a smooth graph for $\hat{x}_c(t)$, choose at least 20 points per sampling interval and provide the resulting time vector `t` as the second output parameter.
   (b) Let $x_c(t) = 1$. Plot $x_c(t)$, $\hat{x}_c(t)$, and $e(t)$ for $T = 1$ and $N = 5$, 10, 15, and 20. Explain the shapes of the plotted signals. How does the error behave as a function of t?
   (c) Let the "size" of the error be defined as the maximum value of $|e(t)|$ in the middle third of the plotted time interval. Determine and plot the size of error as a function of N for $3 \le N \le 20$. Comment on your results.
   (d) Repeat the previous two parts for $x_c(t) = \cos(2\pi Ft)$ with $F = 1$ and $T = 0.1$. Compare your results with the $x_c(t) = 1$ case.
   (e) Repeat part (d) for various values of F and $T = 1/F_s$. Is the size of the error dependent on the normalized frequency $f = F/F_s$?
   (f) Now suppose $x_c(t) = \cos(2\pi Ft + \theta)$ with $F = 1$ and $T = 0.1$, and consider various values of θ. Do the results depend on θ?

11. **Computer assignment** Consider the first-order system described by $y[n] = 0.8\,y[n-1] + 0.2\,x[n]$. It is excited by by the *linear FM signal* given by

$$x[n] = \cos\left[\pi\left(\frac{B}{F_s N}\right)n^2\right] \qquad 0 \le n \le N$$

where $B = 10$ Hz, $F_s = 100$ Hz, and $N = 1000$.
   (a) Determine and plot $|H(e^{j2\pi F})|$ over $0 \le F \le B$ Hz.
   (b) Plot $x[n] = x(nT_s)$ over $0 \le t \le \tau$ sec.
   (c) Process the input $x[n]$ through the system using MATLAB to obtain $y[n] = y(nT_s)$ and plot it over $0 \le t \le \tau$ sec. Verify that the amplitude of $x[n]$ is attenuated according to the frequency response $H(e^{j2\pi F})$.

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EE2004: Digital Signal Processing

Tutorial 11
(April 27, 2026)

1. Let $x[\cdot] = \{\overset{\downarrow}{3}, 2, 3\}$. Find the 3-point DFT of $x[n]$.

2. Let $x[\cdot] = \{\overset{\downarrow}{3}, 0, -2, 2\}$. Find the 4-point DFT of $x[n]$.

3. The 8-point DFT of a real-valued signal $x[n]$ is known over $0 \leq k \leq 4$ to be $X[k] = 1 + 2j\,\delta[k-2] + \delta[k-4]$.
   (a) Determine $X[k]$ for $5 \leq k \leq 7$.
   (b) Determine $x[n]$.

4. Consider a complex signal consisting of two closely spaced sinusoids, i.e., $x[n] = e^{j2\pi(0.30)n} + e^{j2\pi(0.33)n}$. For each of the cases below, plot the $|X[k]|$ as a function of $f_k = k/N$.
   (a) Compute and plot the 10-point DFT of $x[n]$ using 10 samples ($0 \leq n \leq 9$). From the plot, can both exponentials be identified? Explain.
   (b) Zero-pad the signal from (a) with 490 zeros and computed the 500-point DFT. Does this improve the picture of the DFT. Explain.
   (c) Compute and plot the 100-point DFT of $x[n]$ using 100 samples ($0 \leq n \leq 99$). From the plot, can both exponentials be identified? Explain.
   (d) Zero-pad the signal from (a) with 400 zeros and computed the 500-point DFT. Does this improve the picture of the DFT. Explain.

5. Repeat the previous problem when $x[n] = e^{j2\pi(0.300)n} + e^{j2\pi(0.315)n}$

6. Consider the length-4 unit amplitude pulse $x[n] = u[n] - u[n-4]$.
   (a) Compute and plot the 4-point DFT $X[k]$ of the signal $x[n]$.
   (b) Use the DFT interpolation formula to compute the DTFT $X(e^{j\omega})$ from the DFT $X[k]$. Sample this $X(e^{j\omega})$ over $0 \leq \omega \leq 2\pi$ using 10 equally spaced points to create $X_{10}[k]$. Compute and plot the IDFT of $X_{10}[k]$. Comment on your results.

7. Using only the properties of the DFT and the fact that $\delta[n] \longleftrightarrow 1$, determine the 4-point DFTs of the following sequences: (a) $x_a[\cdot] = \{\overset{\downarrow}{0}, 2, 0, 0\}$, (b) $x_b[\cdot] = \{\overset{\downarrow}{0}, 0, 0, 4\}$, (c) $x_c[\cdot] = \{\overset{\downarrow}{0}, 0, j, 0\}$, (d) $x_d[\cdot] = \{\overset{\downarrow}{0}, 4, -j, 2\}$.

8. Find $x_1[n] \circledast_4 x_2[n]$ where $x_1[\cdot] = \{\overset{\downarrow}{1}, -2, 3, -4\}$ and $x_2[\cdot] = \{\overset{\downarrow}{1}, -1, -2, 2\}$.

9. For the previous problem what is the result of linear convolution? For what value of N do the results of circular convolution and linear convolution match over $0 \leq n \leq N-1$.

10. Let $x_c(t) = \cos(2\pi F_0 t)$ where $F_0 = 1\,\text{kHz}$. This signal is sampled at a frequency $F_s = 8\,\text{kHz}$. Let $x[n]$ denote the samples of $x_c(t)$ taken over two cycles.
    (a) Compute the 16-point DFT and plot the magnitude (stem plot). In these and other plots, the x-axis must be in Hz over the range 0 to 4000 Hz.
    (b) Repeat the above part for the following lengths (after zero-padding): 32 and 64. Overlay these results (using different colours) on the stem plot for the 16-point DFT.
    (c) Now zero-pad $x[n]$ such that the total length is 8192. Overlay the normal magnitude plot (don't use stem plot for this part) on the previous plots. Are you able to see that you can view the DFT as the samples of the DTFT?

11. Let $x[n]$ be the same sequence that was defined in the previous problem.
    (a) Suppose we want to consider, as before, just two cycles of $x[n]$. Over what range of values n should $x[n]$ be defined if you want the DFT to be real-valued? How should these samples be arranged to form the vector x for the MATLAB command `X = fft(x)`. Verify that the DFT coefficients are indeed real-valued (or, at best, the imaginary parts are negligibly small, caused by round-off errors).
    (b) Repeat the previous part for $N = 32$ and $N = 64$. That is, $x[n]$ continues to be made of two cycles, but zero-padded appropriately to get the 32-point and 64-point DFTs. We require the DFTs to be real-valued for each of these lengths. How should the samples of $x[n]$ be arranged in each case? ◇